Nimbus: Architecture and design choices

Nimbus project documentation

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Nimbus separates the atmospheric calculation from the application used to configure and inspect it. The simulator is a C++17 executable, independently developed rather than ported or forked from CM1. Nimbus Studio is a Tauri desktop application: TypeScript and React provide the interface, while Rust owns filesystem access, process control and native data readers. Studio launches the simulator as a separate process. The same executable can run from a terminal or on a remote Linux computer without an open Studio window.

This boundary makes a saved configuration and the simulator that interpreted it the starting point for reproducibility. A plot is a view of a completed simulation product, rather than part of the atmospheric time integration. The mathematics guide explains the physical and numerical assumptions; this guide explains where those assumptions are enforced and who owns the data. The execution chapters follow configuration, host storage, CPU portability, GPU submission and numerical precision. The publication chapters then trace persistence and IPC; the rendering chapter follows those published values through shaders to pixels. The Studio guide expands the application workflows, plot interpretation, resource controls, Python and figure-publication contracts. The final chapters separate performance models, historical observations and the evidence required for a new change.

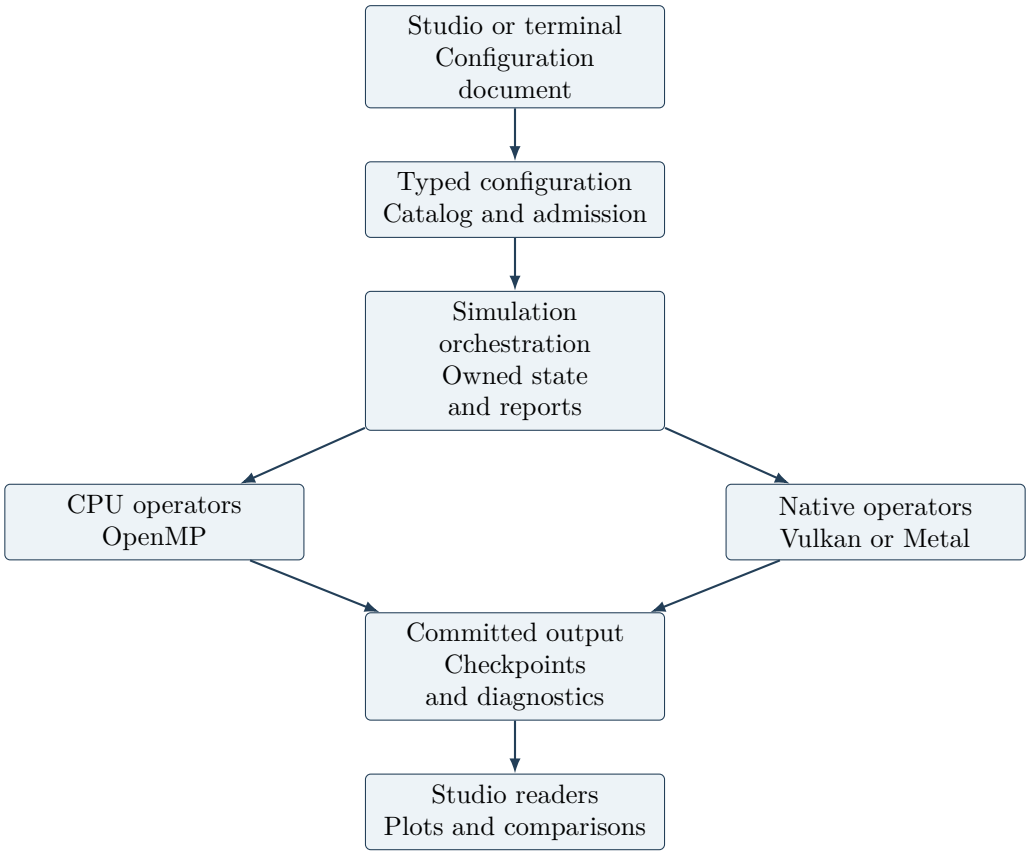


Figure 1: The application and solver share configuration and publication contracts. The numerical state remains owned by the solver. Backend-specific memory and synchronization operate inside that ownership boundary.

1 Local experimentation and the decision to use HPC

A central goal is high-fidelity atmospheric simulation on ordinary workstations and MacBooks through efficient algorithms, careful memory ownership and GPU acceleration. Local work is not intended to stop at toy models, coarse previews or preliminary checks before an inevitable cluster run. Reducing avoidable work and data movement can make substantially more demanding experiments practical on the same hardware. A researcher should be able to change a documented experiment, inspect its assumptions, and increase its resolution without maintaining a separate solver fork for each case. The objective is useful scientific work per unit of time, memory and effort. A high GPU utilization percentage is not sufficient evidence of that objective, and a run does not become scientifically adequate merely because it completes on a laptop or a cluster.

Three questions determine where an experiment belongs. First, does the discretization and selected physics answer the scientific question? Second, does the complete working set fit the available resources? Third, can the required duration, repetitions and analysis finish within the researcher's time budget? A configuration can pass one question and fail another. The mathematics guide's resolution and experiment-design chapter develops the first question and the scaling assumptions behind the other two. This section describes the engineering boundaries that keep the answers separate.

1.1 Capacity and turnaround are different constraints

A useful memory estimate follows simultaneously live owners. It includes primary fields, time-integration stages, scheme-private state, geometry, rollback candidates, reports, output snapshots and communication buffers. The simulator, Studio, Python and operating system can all be active at once. On a discrete-memory machine, RAM and each GPU's local memory are separate constraints. On a unified-memory machine, competing allocations can consume one physical capacity even though several subsystems each report a plausible local budget. Neither topology removes maximum allocation sizes or the need for headroom.

For illustration, a float32 scalar array of size 512^3 occupies 0.5 GiB. The existing 51-field-equivalent dynamics admission envelope would account for 25.5 GiB for that allocation shape before interpreting additional owners and host/device duplication. Doubling all three dimensions makes a scalar array 4 GiB and that envelope 204 GiB. These are arithmetic examples, not recommended Nimbus grids, measured peak usage, or predictions that a device with that nominal capacity will admit the job. The memory admission owner must also check the largest buffer, backend capabilities and simultaneous levels. A preflight estimate should explain which owners it covers; it cannot serve as a measurement of peak process memory.

Turnaround includes more than stepping. A useful comparison is

$$T_{\text{local}} = T_{\text{prepare}} + T_{\text{solve}} + T_{\text{publish}} + T_{\text{analyse}},$$

with overlapping phases counted on the actual critical path rather than added twice. Remote execution also introduces queueing, staging, result transfer and recovery costs. A cluster can shorten the solve and still delay the answer to a small experiment. Conversely, an overnight local run may be unsuitable when the scientific conclusion needs dozens of realizations. Measure the whole workflow for the same experiment before choosing placement.

1.2 Choose the kind of parallelism that serves the question

Need	Useful execution arrangement	Consequence
One admitted case fits locally	CPU or one selected accelerator	Fast iteration can avoid remote staging; actual elapsed time still needs measurement.
Many independent cases fit individually	Separate processes or batch jobs	Distribute experiments without halo exchange; each job still needs a complete working set and a distinct output identity.
One case exceeds a device's memory	Spatial domain decomposition	Requires the distributed ownership and exchange design in the multi-GPU section; adding cards alone does not implement it.
One case fits but misses the time budget	Profile first, then consider strong scaling	Transfers, reductions and output can erase a faster kernel or the benefit of an additional device.
The ensemble or archive exceeds local capacity	Managed remote resources	Scheduling, storage, transfer and reproducible restart become part of the experiment, not administrative details.

Running independent cases is often the first useful step beyond one machine. They can share immutable coefficient files but must not share writable run directories, mutable model state or a GPU allocation budget without explicit coordination. Device assignment and the number of simultaneous jobs must be controlled. The current remote-compute workflow launches a process through SSH; it does not provide a distributed solver or a general cluster scheduler. An ensemble diagram in Studio is an analysis facility, not evidence that a multi-node ensemble scheduler exists.

Local alternatives include a shorter diagnostic experiment, a different output cadence, a smaller scientifically justified domain, or an admitted nest focused on the region of interest. Each changes a different cost. A shorter duration cannot answer a question about late-time behavior. Smaller domains change boundary influence. A coarser grid changes resolved scales. Nesting introduces exchange and feedback requirements and does not bypass their readiness gates. These choices must remain visible in the experiment record rather than being silently selected by an automatic speed policy.

1.3 What optimization can buy without changing the experiment

Memory layout, traversal order and cache reuse can reduce the cost of moving the same physical state. Reusing valid material coefficients can avoid recomputing them at every acoustic operation. Keeping intermediate state on the accelerator can avoid unnecessary host round trips. Reusing scratch capacity can reduce allocation overhead, while shortening actual buffer lifetimes can reduce peak memory. These improvements answer different bottlenecks and can combine to make a demanding local experiment practical. They must retain the producer/consumer, stage and failure contracts developed in the execution chapters.

GPU acceleration is most useful when enough suitable work can remain on the device and the selected numerical families have native implementations. Launching many tiny kernels or repeatedly copying complete fields through the host can consume the anticipated gain. Conversely, a small

diagnostic or control task can be inexpensive on the CPU; moving every function to a shader is not itself the objective. Measure which owner limits the complete run, then remove that cost without silently changing the selected physics, output or acceptance conditions.

The result can be invested in finer resolution, longer duration, more ensemble members or faster interactive iteration. The mathematics guide explains why linear-resolution gains are nonlinear in runtime, and why memory and time must both be considered. A faster implementation and a smaller live working set can shift the practical local/HPC boundary. That boundary should be re-measured after optimization rather than inferred from the resource demands of an early implementation.

1.4 Changing an experiment without rewriting its infrastructure

Existing public choices should be expressed through the versioned configuration and recorded with the executable catalog. Initial conditions, physical domain, boundary conditions, admitted physics, output and backend selection belong to that document. A case-specific condition hidden in an inner loop cannot be recovered by reading the saved YAML and is therefore a poor extension boundary.

A new physical formulation still requires implementation. The reusable unit is its state and operator contract: the quantities and units it reads, the inventories it changes, grid locations, source interval, private continuation state, invalid-input behavior and supported backends. It must join ordinary configuration loading, admission, restart, diagnostics and tests. If it requires native execution, a host implementation alone does not complete the feature. The existing typed backend interface and scheme catalog provide boundaries to extend; they are not a dynamic plugin system that can accept arbitrary new physics without compiled code.

This also separates analysis customization from solver customization. A new Python diagnostic can operate on published fields without changing the time integrator. It must disclose missing inputs, units, grid reconstruction and sampling. A new prognostic variable has a much larger obligation: initialization, transport, source coupling, halos, restart and publication all need an owner. Modern engineering reduces the repeated plumbing, but does not make those scientific obligations optional.

1.5 Measure a decision boundary, not a universal machine ranking

A useful future resource report would combine observed peak working sets, stage times, transfers, output volume and actual simulated duration with the configuration identity. It should state uncertainty and identify extrapolation. The existing admission checks, telemetry and Studio loading-worker measurement cover pieces of this problem; they do not yet form a universal experiment-cost estimator. In particular, Studio's small cached NPY measurement chooses a loading heuristic and cannot choose solver resolution or predict a storm's runtime.

The practical development order is to establish correctness for a declared profile, optimize and measure complete single-device work on representative unified and discrete-memory systems, then evaluate independent-job throughput and finally coupled distributed execution where the experiment requires it. Lowering the barrier means keeping the local build usable throughout that process. Optional cluster transports should not become prerequisites for running a single CPU case, and a proposed scaling feature should not be advertised as available before it has public configuration, working coupling and the required tests.

2 The solver module map

The execution machinery described later serves a set of physical and numerical modules. Their boundaries are easiest to understand by following what each consumes, what it produces, and which state must survive to the next call. The following map covers the top-level solver directories under `include/` and `src/`. It is an ownership index, not an assertion that every factory entry or combination is Ready. The current catalog and readiness registry remain authoritative for admission.

Module and interfaces	Implementation responsibility
Core interfaces	Core implementation: field storage, physical geometry, configuration/runtime state, headless lifecycle, nested execution, checkpoint and output owners; orchestration composes accepted operator changes.
Initialization	Typed configuration and initial sources: sounding-column construction, hodograph selection, trigger selection and requested/realized environment diagnostics.
Sounding readers	File ingestion: auto routing, CM1 and SHARPY-related readers, unit conversion, profile quality control and interpolation.
Dynamics	Cartesian A-grid/C-grid tendencies, reference-state forces, native boundaries, conservative momentum and moving-frame owners.
Numerics	Advection, diffusion and integration: TVD/WENO, retained carrier fluxes, explicit/implicit diffusion, split-explicit/RK methods and their stability contracts.
Boundary conditions	Geometric closure: staggering-specific boundary execution, closure storage and private transported-field boundaries.
Microphysics	Kessler, Lin, Thompson and Milbrandt: water sources, private particle moments, sedimentation, thermal attribution and state commit.
Radiation	Simple-grey and RRTMG: physical columns, optics, fluxes, heating, mapped shortwave transport and held-source energy accounting.
Boundary layer	Slab, YSU and MYNN: column mixing, surface exchange, persistent column state and conserved interval-mean forcing.
Turbulence	Resolved-grid SGS: Smagorinsky variants and prognostic TKE, coefficient fields, stress/scalar fluxes and production budgets.
Terrain	None/Bell/Schär topography, coordinate metrics, physical flux transforms and mapped transport geometry.
Chaos	Controlled stochastic perturbations: keyed random streams, spatial/temporal correlation, source coupling and retained epochs.
Radar	Synthetic observation operators: reflectivity, radial velocity and polarimetric products with declared geometry and particle inputs.
Diagnostics	Field contracts, budgets and derived products, storm tracking, rotation diagnostics and parcel trajectories.
Compute	Backend-neutral runtime, CPU owners, typed dispatch and residency; native GPU implementations live under Vulkan and Metal.
Utilities	Shared scheme registration, numeric text, logging, cadence, transient workspace and copy helpers; relevant compiled helpers also live in core infrastructure.

Module and interfaces	Implementation responsibility
Solver tools	Field-contract validation and Thompson activation/collection/cold-table generation. These executables support provisioning and tests; they are not additional atmospheric schemes.

There is no separate implemented soil or prognostic land-surface directory. The similarly named `surface.scheme` and `boundary_layer.scheme` controls have different responsibilities, explained below. A directory inventory must not invent a model because a related name appears in configuration or in a paper.

Most physical families use the descriptor-backed SchemeRegistry. It validates canonical IDs, aliases and constructor agreement. A constructor establishes that an object can be built; it does not establish valid input geometry, complete private-state continuation, native dispatch or a supported coupling. Initialization sources and geometric boundaries have their own typed factory dispatch. These different factories should not be forced into one lifecycle merely because they all select an implementation.

3 Initialization is a composition of three sources

3.1 Thermodynamic column construction and file reading are separate

`src/init/sounding` owns the model-column source. Its choices are `parametric_cape`, `parametric_targets` and a file-backed source. The source returns a `Sounding` containing physical heights, temperature, theta, dry-air vapor mixing ratio, pressure, dry density and optional Cartesian winds. The caller broadcasts or physically samples that column onto the selected grid. The source is not simply a filename reader. See the column interface and source factory.

The separate `soundings` module supplies the auto, CM1 and SHARPY-related file readers. It interprets text and supported netCDF/HDF5 representations, checks required variables and units, and interpolates to requested heights. Strict file initialization requires adequate vertical coverage and rejects missing or invalid inputs rather than silently creating an unrelated profile. Lower-level compatibility code contains explicit fallback options; their existence does not change the ordinary strict-v1 no-fallback contract.

The file-column adapter reintegrates pressure on the model column. A file pressure at the first interpolated scalar center is not automatically a surface-face pressure; without that supplied center pressure, the surface boundary must be integrated to the first center. Density and theta then follow the retained dry thermodynamic convention. This separates observed input representation from the reference state the discrete force operator actually consumes. It does not prove exact hydrostatic balance for every possible later stencil; the initialization and resting-column tests check the implemented pairing.

Parametric targets similarly produce a realized column and diagnostics. Requesting CAPE, humidity or a characteristic level does not make every environmental diagnostic independently controllable. Requested and realized values must remain distinct. A parameter solver's failed or out-of-domain target should not be reported as an exact physical initial condition just because it returned a vector of the requested length.

3.2 Wind selection and perturbations have explicit precedence

The hodograph factory handles `auto`, `wk_param` and `zero`. Auto uses complete winds from the sounding when available; otherwise it builds the retained parametric hodograph. Explicit parametric or zero selection has its own meaning rather than being silently replaced by file winds. The resulting Cartesian wind column is placed on the chosen scalar/face locations by initialization orchestration. Wind direction interpolation should operate through components where required, avoiding a spurious angular jump across north.

The trigger factory selects the warm bubble, no perturbation or a vortex seed after the base column and wind state exist. The current Cartesian warm-bubble owner is not theta-only: for positive heating with available pressure/vapor fields, it rescales vapor using the ratio of its retained saturation estimates at the old and new temperatures. Optional cloud-liquid seeding is another declared increment. This is an imposed initial-state perturbation, not a closed microphysical condensation transfer. Its moisture and thermal assumptions must be included in reproducing an experiment.

The current vortex seed adds a horizontal Rankine profile about its x/y center up to the hard `z_top_m` cutoff. Its public `center_z_m` parameter is parsed and serialized but currently has no effect in the seed application. The owner does not author a balancing pressure field, and its discrete staggered field must not be described as a proven balanced or divergence-free vortex. These are implementation limits to retain when interpreting the seed source.

Terrain initialization uses physical sample heights, and subsequent reference and EOS reconciliation belongs to orchestration. Trigger code should not be treated as an independent complete thermodynamic integrator. The permanent file-source, target-column, hodograph and trigger tests define bounded checks at these interfaces.

4 Resolved dynamics, numerical operators and boundaries

The dynamics factory constructs distinct Cartesian A-grid and C-grid schemes. Their tendencies consume the state view, base profiles and physical geometry; the orchestration layer owns applying them in a valid sequence. Pressure gradients, gravity, vapor buoyancy, condensate loading, rotation and advection are not interchangeable source plugins with arbitrary ordering. A native operator must implement the selected grid's actual field locations and force convention.

`numerics` owns the reusable spatial and temporal algorithms. Its advection factory provides TVD, WENO5 and disabled transport; explicit and implicit diffusion have separate operators and configured coefficient sources. Time integration selects split-explicit, SSPRK3 or classical RK4. A disabled configuration may bypass an operator rather than construct a meaningful diffusion solve with zeroed input. Interface-flux registers retain the mass carrier and interval needed by scalar transport and nested feedback.

The distinction between a tendency and a committed state matters here. An RK stage can require primary fields, private microphysics state and reports to be staged together; a split acoustic callback has different pressure/boundary dependencies. The dry rho–theta EOS projection diagnoses pressure from stored density and theta with its declared arithmetic. It does not become a moist gas-law solve because a neighboring physics source includes water energy. Numerical diffusion, SGS stress and PBL mixing also remain separate owners; selecting one does not implicitly remove the others' coupling constraints.

Boundary construction dispatches on coordinate system and staggering, while its runtime chooses the level's execution object. Physical walls, periodic connections and open/radiative conditions

define closure values and histories needed by later stencils. They do not turn halo storage into physical atmospheric volume. Private transported moments require corresponding boundary closure and cannot inherit the primary water field's state by accident.

Surface exchange is not geometric boundary filling. A rigid lower normal-flow condition can coexist with a surface stress or heat flux owned by the PBL coupling. Conversely, copying a temperature into a ghost layer does not automatically supply a physically accounted surface energy flux. Native boundary, source and diagnostic owners must agree on that separation. See boundary runtime, numerics runtime and step orchestration.

5 Microphysics owns more than the visible water arrays

The microphysics registry supplies Kessler, Lin, Thompson, Milbrandt and disabled physics. The math guide describes their different process and spectrum assumptions. Architecturally, the shared interface accepts pressure, theta, live dry density, water species, physical layer information and the interval, and produces primary changes plus any scheme-owned moment changes and receipts.

Thompson and Milbrandt retain private fields that must follow transport, boundary closure, cloning, relocation and checkpoint continuation. The base contract distinguishes parameter cloning from evolved state, exposes the storage units of transported private scalars, and separates preparation/validation from publication. An accepted q update with old N or Z would describe a different spectrum even if total water still summed correctly.

The microphysics step owner integrates those results with primary-state acceptance and budgets. Sedimentation is a physical lower-boundary exchange with a precipitation receipt, not merely another local category conversion. Execution must reach the process and commit owners declared by admission: an explicitly host-owned commit is different from silently replacing a required device operation. A device capability bit alone proves neither. Ordinary double-moment Milbrandt and the guarded triple-moment work are separate scopes; the latter's bounded wet-surface/EOS helpers do not complete public field-level admission.

6 Radiation: columns, spectral transfer and held power

6.1 The column boundary makes physical inputs explicit

The radiation interface defines temperature, pressure, dry density, physical layer thicknesses and interface heights, optional interface pressure/temperature, water profiles, gas composition, solar geometry and surface optical/thermal state. Center pressure is not an interchangeable substitute for pressure at the surface or top interface. Likewise, nominal grid spacing cannot replace mapped column thickness. Column evaluation checks dimensions, finite input, flux/tendency shape and the declared column-power closure before publication.

The factory contains `none`, `simple_grey` and `rrtmg`. Omitted radiation means no radiative forcing, not an automatically inserted clear-sky model. Simple-grey offers a pressure-based clear-sky optical profile and a distinct absorbing vapor/liquid/ice-path model. The latter's pure absorption assumption does not represent cloud scattering. The collimated incident solar beam and Lambertian surface-reflected flux have different angular meanings; sharing a broadband coefficient does not make them the same ray distribution.

RRTMG is a C++ spectral implementation whose AER coefficient files are explicit scientific inputs. Coefficient loading, atmosphere conversion, reference-grid interpolation, gas/cloud optics and long-

wave/shortwave transfer have separate source owners under the RRTMG directory. It does not call a reference Fortran program at runtime. Its resolved-condensate cloud optics represent a fully cloudy cell where applicable rather than inventing a subgrid cloud-fraction/overlap model. Explicit gas mixing ratios and particle effective radii are part of the closure, not hidden renderer settings. Dry-gas profiles use dry-air mole/volume fractions, while water inputs retain their declared dry-air mass ratios; the atmosphere adapter owns the conversion to molecular columns. Missing coefficients or unsupported inputs must not turn RRTMG into simple-grey.

6.2 Heating and refresh cadence have separate ownership

Radiative fluxes produce a column temperature tendency through the retained physical mass and heat-capacity convention. That tendency is converted to theta for coupling, while conserved power and flux diagnostics are published with it. Radiation does not directly replace the dry EOS. Agreement of its column power with integrated heating checks one source boundary; the later thermal and pressure commits still need their own energy attribution.

Radiation runtime retains last/next sample times and holds the accepted source between refreshes. It constructs candidate column fields and reports, rejects failed columns before publishing the new generation, and retains the previous accepted source when a refresh fails. The cadence is part of continuation state. Recomputing an otherwise held source immediately after restart can change subsequent evolution even if every visible field initially matches.

The application ledger compares held atmospheric power times the applied interval against booked radiation heat. Mapped shortwave also distinguishes incoming power, atmospheric and surface absorption, top and lateral escape, and unresolved reflected power. Reflection convergence is an acceptance condition, not a number to hide inside the atmospheric tendency. These receipts describe actual source publication and application; they do not by themselves prove the complete model conserves a physical total energy.

6.3 Terrain and native execution are additional contracts

Terrain shortwave uses slope/visibility, physical interfaces, lateral boundary semantics and angular/reflection work beyond independent vertical columns. The current CPU mapped RRTMG path has a spectral terrain operator. Its radiation runtime explicitly rejects mapped RRTMG shortwave on an active accelerator because the required native spectral-terrain owner is missing. This limitation must not be inferred away from the existence of RRTMG native column batches.

For supported column work, the runtime asks the scheme whether the selected backend supplies the required native batch operations. Rejected or unavailable dispatch fails a no-fallback profile; CPU fallback is used only when explicitly permitted. Geometry and output-diagnostic combinations can impose further restrictions. The term “C++ implementation” therefore does not mean every spectral branch is device-native. See batch publication, radiation references and runtime-state checks.

7 SGS, boundary-layer and surface exchange are different models

7.1 Resolved-grid SGS

`turbulence` constructs Smagorinsky, prognostic TKE or disabled SGS. Smagorinsky variants own coefficient/filter/stability choices; the TKE family adds an evolved energy field, mixing length,

shear and buoyancy production, dissipation and transport. Their spatial stress and scalar fluxes must respect the chosen staggering and physical metrics. Returning an eddy viscosity field is not the complete operator: application to momentum/scalars, realized energy change and boundary work remain separate steps.

SGS runtime and SGS application connect closure preparation to accepted changes. C-grid tensor placement, collocated derivatives and mapped stresses have different contracts. Dynamic Smagorinsky's bounded coefficient and averaging policy do not automatically authorize a vertical-only variant; the retained static vertical-only scope is still tracked separately as Engineering incomplete. A filter width is a physical closure parameter even when its default is derived from the grid.

7.2 Column PBL and interval-mean forcing

`boundary_layer` constructs slab, YSU, MYNN or disabled mixing. Slab retains per-column mixed-layer depth and thermodynamic state; YSU and MYNN have their own nonlocal/stability/TKE assumptions. Their column workspace receives physical centers/interfaces and scalar-location wind reconstructed from the actual grid. A PBL column is not a view of all allocated halo entries.

Column evaluation checks completed results before runtime publication. The held representation includes forces and scalar inventory rates, not just copied accelerations whose denominator may later change. Applying a retained force to the current face mass requires the appropriate finite-step momentum/energy bookkeeping. Implicit column exchange publishes its realized interval-mean fluxes; requested diffusivities alone do not describe that accepted transfer. Persistent column state, cadence and held forcing belong in rollback and restart.

Native batch submission is another owner, with bounded packing and complete publication. Its presence does not license all cloud-top, terrain or SGS/PBL compositions. Admission must prevent two closures from unintentionally modeling the same unresolved transport twice. See PBL reference checks and PBL state checks.

7.3 Surface state does not imply a soil model

The public surface choices are `none` and `slab`. Here surface selection controls lower-boundary exchange with an active PBL closure; it is distinct from selecting the slab PBL family. Monin–Obukhov similarity and related surface exchange live in the PBL implementation, including surface-flux helpers. `SurfaceConfig` carries roughness, albedo, emissivity, skin temperature and surface vapor mixing ratio. If the prescribed skin values are unset, the coupling captures the initial surface-air values and holds them as the surface reference.

There is no implemented multilayer soil temperature/moisture, vegetation or prognostic land-water reservoir implied by that interface. Surface temperature and moisture are prescribed/held forcing for the atmospheric exchange, not the endpoint of an unshown soil energy equation. Radiation's surface absorption receipt likewise does not automatically evolve skin temperature. Adding a land model would require new state, an energy/water exchange contract, coupled time integration, restart and admission rather than relabeling existing slab controls.

8 Controlled stochasticity, synthetic radar and diagnostics

8.1 Chaos is a perturbation owner

The chaos registry provides disabled, initial-condition, boundary-layer and full-stochastic families. These introduce controlled variability through declared perturbation interfaces; `boundary_layer` chaos does not replace the selected physical PBL scheme. Keyed random streams, correlation filters, member/seed identity and retained epochs belong to the experiment. The random-stream implementation and runtime state prevent one logical request or rejected update from silently becoming another request's history.

Moving-domain noise also has a declared Earth/frame geometry. Reusing an array index after relocation is not a physical coordinate definition. Source perturbations must preserve their underlying inventory/flux contract; scaling a completed sedimentation divergence independently in every cell would not preserve its surface precipitation exchange. Reproducibility claims remain limited to the tested generator, arithmetic and continuation scope; a seed alone is not a cross-library bitwise guarantee.

8.2 Radar is a forward operator, not microphysics state

The radar factory selects reflectivity, velocity and polarimetric ZDR operators; product selection also exposes associated KDP and correlation outputs. RadarStateView binds the chosen mass and number inputs, dry density, pressure/theta, staggering, physical geometry and Earth-frame radar location. Multiplying a dry-air mixing ratio by dry density supplies species content per volume; treating it directly as volume concentration would change the scattering calculation.

Radial velocity reconstructs face winds to the declared sample location before line-of-sight projection and any configured scatterer fall-speed correction. Terrain heights and grid origin matter to the beam direction. Reflectivity and polarimetric tiers make particle/scattering assumptions beyond the mass fields themselves; a microphysics sixth diameter moment is not automatically every radar product. Radar runtime constructs only selected products and retains their operator state. These are diagnostics of an accepted atmosphere, not additional forces feeding the dynamical state.

8.3 Diagnostics supply contracts, analysis and receipts

Diagnostics owns field membership/units, dry-mass and moisture inventories, momentum and versioned energy, resolved derivatives, pressure decomposition, water paths, storm tracking/rotation and parcel trajectories. Its diagnostic pressure decomposition must not be confused with the rho–theta EOS pressure commit. Its physical-domain sums must not include closure volumes merely because those arrays contain finite values.

Some outputs require state across time: accumulated rainfall, parcel paths, storm continuity and moving-window histories cannot be reconstructed from one instantaneous field snapshot. Other products are derived lazily from an accepted endpoint. Product registration connects their dependencies to output selection, while field contracts describe interpretation and bounds. Optional plotting products and required scientific ledgers have different cadences and obligations.

An energy receipt attributes a change using a declared inventory; it does not prove the corresponding physical heat or work law. Storm tracking is likewise an algorithmic interpretation of fields with continuity and ambiguity rules, not a uniquely identified physical object. The maintained diagnostic tests exercise those definitions and their finite-coverage/failure conditions.

9 Terrain, nesting and the common lifecycle

The terrain factory provides flat, Bell and Schär geometry. Its physical heights, interfaces, Jacobians and metric signs feed dynamics, transport, column physics, diagnostics and output. Terrain changes numerical geometry; it is not an actor transform applied only when drawing. Grid-attached and Earth-fixed placement also have distinct motion semantics.

Nesting is coordinated within core infrastructure rather than by one independent "nesting physics" factory. Level runtimes own their schemes and contexts; boundary fills, subcycling, restriction and reflux connect parent and child inventories. Relocation/recycling additionally owns newly exposed regions, private moments, held sources, physical origins and reports. The relevant boundaries are nested execution, scalar feedback and recycling state. Exact stationary and moving reference families are distinct from the broader Engineering-incomplete moist/moving nesting item.

The headless runtime brings these modules together: initialization and admission create the state; cadence owners refresh held physics; dynamics/orchestration advances it; diagnostics and output describe committed endpoints; checkpointing retains the full continuation state. Compute interfaces implement particular operations inside that lifecycle, while utility helpers provide common parsing, cadence, copy and registration rules. None replaces the higher-level physical owner.

Finally, `src/tools` contains table-generation and validation programs. Generated coefficient/table files have explicit provisioning and identity requirements. Their absence cannot be repaired by substituting a different closure. These support tools, the included interfaces and the implementation tree form a maintainable module map; long campaigns and independent review remain separate evidence stages for the exact scientific combinations.

10 Configuration, executable identity and admission

The simulator accepts a document, but it executes a compiled plan. The YAML reader first checks syntax, field names, types and ranges. Typed configuration then identifies the selected grid, schemes, numerical parameters, output policy and backend. The configuration compiler resolves the capabilities and couplings those choices require. Finally, ordinary runtime admission decides whether this particular combination can execute. These are separate questions: a number can be syntactically valid yet lie outside an implemented physical domain, and two individually available schemes can require an unfinished coupling when selected together.

The authoritative path is visible in the typed loader, configuration compiler, readiness registry and executable entry point. The registry distinguishes Engineering incomplete from Ready. The runtime word `qualified` selects an admission policy; it does not assert that a long campaign or external scientific review has occurred. Public Milbrandt triple-moment selection remains guarded while its general source and live thermal coupling are incomplete.

Studio does not maintain a second independent scientific model of which configurations should work. Its frontend uses the executable's scheme catalog, and its Rust preflight layer checks the source document and executable identity again before launch. A browser snapshot provides the same catalog structure when no native request is being made, but that snapshot is a generated input with an authority fingerprint. An old snapshot must not silently authorize a new executable. Conversely, a valid old run does not acquire new capabilities merely because the currently installed Studio knows additional schemes.

Some narrow admission scopes bind exact canonical configuration digests. This is stricter than

matching a family name such as Metal or ZFP. The strict serializer fixes the field inventory, defaults and representation; a membership test then checks that canonical document against the declared scope. Separate manifests bind raw source identity, serialized output-block identity and the combined admission scope. Those hashes answer different questions. Formatting the serializer can change its source hash without changing its serialized block; changing the source-bound manifest then changes the hashes of downstream combined manifests. Updating a block digest to stand in for a manifest digest would be an identity error even if both strings are valid SHA-256 values.

The C++ catalog, TypeScript validator and Rust preflight therefore have coordinated exact-scope checks. See catalog generation, frontend validation and native preflight. These checks are not a substitute for numerical tests; they keep the tested scope, selectable scope and user-visible scope from drifting apart. A metadata refresh must preserve every non-hash field unless a separately justified scope change is intended.

11 Field storage, layout and ownership

11.1 Contiguous fields and the cost of a traversal

Field3D owns a contiguous array of binary32 values. Although compatibility accessor names retain historical radial/angular terminology, the current atmospheric configuration is Cartesian. For dimensions (n_x, n_y, n_z) the scalar address is

$$\mathrm{offset}(i, j, k) = (in_y + j)n_z + k.$$

The vertical index is contiguous. A vertical column sweep therefore visits neighboring storage, while a horizontal stencil steps by a column or a slab. This layout favors several column processes and permits simple full-volume loops. It also means that a horizontal and a vertical stencil have different cache behavior even on an isotropic physical grid.

State is organized as separate fields rather than an array of large cell objects. A pressure operator can read density, theta and the required winds without loading every microphysical number, radiation tendency or diagnostic member attached to a cell. The cost is that an operator needing many fields streams several arrays. For N stored cells and m complete Float32 fields, the payload alone is $4mN$ bytes, before padding, geometry, snapshots, private state, device copies or output. At $128 \times 128 \times 180$ cells, one field is 11.25 MiB. A seemingly minor extra read-and-write pass over ten such fields represents approximately 225 MiB of logical traffic, even if caches or hardware write policies change the actual memory-bus traffic.

The allocator aligns the base to 16 KiB and rounds the allocation size to that alignment. This supports relevant host-memory import requirements and also provides a well-aligned base for CPU access. It does not make every row aligned to every SIMD width: the row stride still depends on n_z. Nor does alignment prove that a Vulkan device accepts importing that allocation. Device memory requirements, supported external-memory features, lifetime and synchronization must still be checked. A successful aligned host allocation is only the first part of such a contract.

Field dimensions are checked before allocation; copying a Field3D copies its payload, while moving transfers its ownership. Raw pointer views do not extend that lifetime. Aliasing is consequently an explicit operator concern. Some helpers permit exact in-place input/output aliases, whereas shifted overlap would overwrite unread input. A `restrict`-qualified vector loop is only valid when its caller satisfies the non-aliasing promise. SIMD speed does not justify relying on accidental overlap behavior. The SIMD alias tests exercise these distinctions.

11.2 Temporary capacity is reusable; contents are not automatically valid

The field pool groups reusable buffers by shape. A normal acquisition clears recycled contents. An overwrite acquisition skips that fill only because its caller promises to write the complete required contents before reading them. Newly allocated vector storage is still initialized by the container, so this optimization removes repeated fills, not every allocation-time write. A poison mode fills overwrite buffers with NaNs to expose incomplete initialization in tests.

The pool protects its bookkeeping with a mutex, but performs full-field fills and copies after releasing that lock. This keeps independent shape requests from serializing behind a large memory traversal. RAII guards return buffers on normal exit or exception unwinding. A cached buffer is still allocated memory: releasing a guard lowers live ownership, but does not necessarily lower process RSS. Peak memory, current live bytes and retained reusable capacity are different measurements.

This pattern matters especially on unified-memory systems. CPU scratch, resident GPU resources, output snapshots and browser textures can compete for the same physical capacity even though each subsystem reports a reasonable local count. On a discrete system, the same workload can exhaust host RAM and VRAM independently. A memory estimate must count the simultaneous ownership graph, rather than multiply the number of prognostic variables by cell count and assume everything else is negligible.

11.3 Simulation state and nested contexts

SimulationStorageState owns the primary fields and associated runtime state. Views expose related prognostic fields, base-state columns, tendencies, geometry and services to operators. They carry access, not ownership. Compatibility entry points still bind a global simulation instance; the design is not a claim that all legacy code is free of globals. New code should use an explicit state view and the correct level's runtime instead of recovering a backend or geometry through an unrelated global.

A nested child has its own dimensions, physical domain, source history and private scheme state. It cannot borrow a parent's buffer merely because the shape matches. A moving grid also changes the physical interpretation of array locations. Generation changes must invalidate stale geometry-dependent or host/device reuse claims. This is one reason a domain move is a coordinated transaction rather than an array shift applied independently to whichever fields happen to be plotted.

12 A timestep is a sequence of owned state transitions

The dynamics owner connects slow tendencies, acoustic pressure/density and momentum updates, boundary closure, retained transport fluxes, physical sources and the final EOS projection. The mathematics guide derives the equations and their locations. The architectural requirement is that every consumer knows which stage its inputs represent. An old density, a newly projected pressure and newly created cloud water are not one coherent stage merely because their arrays have compatible shapes.

A typical bounded operator has a preparation phase, a candidate result and a publication point. Preparation validates geometry, units, dimensions, aliasing, domains and resources. Candidate work can populate scratch fields or backend buffers. Publication changes the caller's authoritative fields and the associated reports only after the result meets the owner's contract. Failure atomicity is scoped to that transaction: it means that a rejected operator does not leave a half-updated set of owned outputs. It does not promise arbitrary rollback of the entire run or reverse changes made

by an unrelated earlier timestep.

Reports are outputs too. An operation changing momentum can also change kinetic energy, boundary exchange and a diagnostic ledger. Updating the wind while retaining the old report, or publishing a report before checking the wind, creates inconsistent state. Similarly, microphysics primary water and private number moments have a shared acceptance boundary. Checkpoint restart is the separate mechanism for returning to an earlier completed model state.

Conservation diagnostics increase this coupling. A cell-centered internal energy inventory and a C-grid face kinetic inventory need different geometry and material inputs. The pressure-material reuse owner retains immutable cell information while pressure changes; the velocity snapshot helper can complete a snapshot without needlessly reconstructing unchanged thermal parts. These are data-reuse changes with explicit preconditions, not changes to physical equations. The relevant maintained sources include pressure cell energy and energy-budget tests.

The native fused acoustic path and the ordinary strict CPU path need not traverse the same implementation. Profiling a similarly named helper is not proof that the selected runtime calls it. An optimization must first identify the actual admission and call path, then preserve the observable stage sequence on that path. This avoids optimizing a dormant implementation while claiming improvement in the active solver.

13 ARM, x86 and x64 are several portability questions

13.1 Instruction set is distinct from operating system and memory topology

In this guide x64 means x86-64, the 64-bit x86 instruction-set family; ARM64 means AArch64. Neither term specifies the number of cores, vector width, memory channels, cache layout, GPU, operating system or graphics driver. An x86-64 host can use an integrated GPU with shared memory. An ARM64 host can have a discrete accelerator. Selecting a memory policy from the CPU brand or pointer width would conflate independent properties.

A native executable must match the target instruction set and operating-system ABI. Object formats, calling conventions, system APIs and dynamic-library dependencies also matter. A generic x86-64 Linux build is not a Windows binary and is not an ARM64 binary. Replacing host-specific instructions does not automatically make a new library ABI compatible with an older operating system. A portable checkpoint similarly means a defined data format, not permission to load it into an incompatible scientific configuration.

Nimbus's build options use `-march=native -mtune=native` for ordinary source builds. These permit the compiler to exploit the build host. `PORTABLE=1` removes those host-native flags and records `NIMBUS_PORTABLE_BUILD`; `DISTRIBUTION=1` requires that portable choice and rejects conflicting host-targeting flags. The resulting baseline remains the chosen compiler and architecture's target. It is not a universal fat binary containing every ISA variant. GCC explicitly distinguishes instruction-set selection from tuning in its x86 option reference.

On macOS, the build records a deployment target and distribution builds turn unguarded availability into errors. The desktop packaging route targets macOS 12.0 to match the pinned MoltenVK runtime and checks bundled Mach-O deployment metadata. This floor is distinct from a standalone solver's source-build default and does not itself prove installed application or WebView acceptance on macOS 12. Windows and Linux require their own compiler/runtime dependencies and packaging checks. CPU-only, OpenMP-enabled and serial builds also differ in capability. A serial build rejects an explicit worker request greater than one instead of pretending that it created a parallel team.

The maintained platform guide states supported build evidence; source branches for a platform are not evidence that every combination has been exercised there.

13.2 The actual SIMD implementation is narrower than the CPU's feature list

SIMD declarations and vector helpers provide scalar, SSE, AVX and Neon implementations for selected array operations. The implemented paths process four Float32 values for SSE/Neon and eight for AVX. An AVX-512 compiler target deliberately uses the tested eight-lane AVX path in this utility layer; there is no separate sixteen-lane implementation there. Reported active width must describe that choice rather than advertising the maximum feature named by the processor.

The code distinguishes compiled features from runtime host features. GCC/Clang x86 capability queries use compiler builtins; AArch64 exposes the Neon path. That reporting helps detect what is available, but the array implementations are selected by compile-time branches. A runtime feature query cannot make a whole translation unit compiled with unsupported instructions safe, and it cannot create an AVX implementation absent from a generic binary. True multiversioning would require separately compiled or target-attributed implementations and a safe dispatch boundary. It must not be inferred from the existence of a feature-report function.

The core solver also relies on compiler vectorization and OpenMP SIMD loops. Therefore the helper width alone is not a census of all vector instructions in a binary. A particular loop may stay scalar because of control flow, aliasing, gather/scatter access, a function call or an unsupported transformation. The useful evidence is the emitted code or vectorization report for the active owner, accompanied by a correctness test. Wider vector instructions do not guarantee proportional speedup when memory bandwidth or synchronization is the limit.

Arm's Advanced SIMD reference describes fixed-width Neon operations. Nimbus's current helper does not become an SVE or SME implementation merely by running on a newer ARM machine. Likewise, generic x86-64 support and native AVX tuning are distinct tested artifacts. Portable correctness should be checked with an actual baseline object set, not by relinking a new entry point against old native objects.

14 Threads, caches and deterministic budgets

14.1 Worker count is a policy input

Nimbus uses OpenMP for shared-memory CPU execution. The CLI gives an explicit `--threads` request priority and disables dynamic team sizing for that request. Otherwise an existing `OMP_NUM_THREADS` remains authoritative. Without either request, topology detection can prefer a detected performance core class or physical cores on an SMT host, bounded by the runtime's available thread count. It does not impose one hardcoded workstation thread count. See hardware detection and the thread override in the CLI.

The detection code uses platform facilities: Apple performance-level counts, Linux processor/package/core identifiers, and Windows processor topology. These are observations, not a guarantee of exclusive access to those cores. Container CPU sets, scheduler restrictions and an OpenMP runtime's processor group behavior can constrain execution. A topology-derived default is a starting policy rather than an automatic optimum for every mesh and scheme.

SMT adds logical execution contexts sharing core resources. Doubling logical workers does not double floating-point units or memory channels. On a bandwidth-heavy stencil, extra workers may

increase competition; on a smaller or latency-limited workload they may behave differently. Heterogeneous performance and efficiency cores pose another scheduling question: one slow participant can lengthen a barrier wait even if aggregate CPU utilization is high. These effects are reasons to measure, not reasons to forbid a topology.

Nimbus records OpenMP binding and place information but does not install a universal affinity policy. `OMP_PLACES` and `OMP_PROC_BIND` remain runtime controls. The OpenMP places and binding definitions explain that these select where threads may run and how they remain associated with places. A logged request is weaker evidence than the runtime's actual placement; some environments cannot honor a requested policy.

14.2 NUMA is a remaining measurement and placement concern

The inspected implementation does not provide an explicit NUMA allocator, `mbind`/`set_mempolicy` placement policy or automatic cross-socket work decomposition. Host vectors use ordinary allocation, and their initialization and subsequent access interact with the operating system's memory-placement policy. Consequently, a physical-core count alone does not establish that a large array is close to every worker that reads it. It would be inaccurate to describe Nimbus as automatically NUMA-optimized.

For a new multisocket or chiplet host, useful questions are which workers touch which slabs, where pages are placed, which cache levels are shared, and whether the working set saturates local or interconnect bandwidth. Changing first-touch behavior or affinity is a separate candidate requiring matched measurements. The current scalar layout and pool capacity reuse are useful foundations, but they do not answer these topology questions by themselves.

14.3 Reduction order is a numerical algorithm

Floating-point addition is not associative. Dividing a sum among a different number of workers usually changes its grouping, even if each cell contribution is identical. Nimbus's deterministic diagnostic path instead forms fixed horizontal slabs, visits each slab in a prescribed order, and merges slab partials in increasing index with compensated accumulation. OpenMP can assign the independent slabs to different workers without changing that partition.

For an accumulator s and an added value x, compensation retains an estimate of the low-order portion lost when $s + x$ is rounded. The implementation uses a magnitude-dependent correction and returns the main sum plus the correction. This improves the sum while fixed ordering supplies thread-count reproducibility. Compensation alone would not fix an arbitrary parallel reduction tree. The contract in deterministic reductions is explicitly for one build target; it is not a blanket bitwise promise across all libraries, architectures and source expressions.

Regular SIMD reduction helpers can use another grouping. They must not be substituted for a deterministic scientific budget merely because their scalar return type matches. Likewise, comparing two simulations requires identifying whether equality means exact stored bytes, fixed-build repeatability, an operator-specific tolerance, or a physical error bound. These are different tests with different failure meanings.

15 Backend dispatch, kernel ABI and memory admission

The backend interface exposes operation families rather than an untyped function that accepts any buffer list. Family availability, typed dispatch fields and validation define what can execute. An

absent family is unavailable; the owning runtime decides whether that is a permitted fallback or a native-required failure. The backend receives its runtime configuration before initialization so that independent simulation contexts need not recover device selection through globals.

Each shader has a stable kernel identity and a manifest contract covering resource bindings, constants, precision and availability. Host constants use explicit integer widths, Float32 fields, padding and size assertions in the kernel ABI. A C++ compiler's ordinary object layout is not automatically a shader language's uniform or push-constant layout. The explicit structures and manifest checks prevent an apparently successful dispatch from interpreting an integer dimension as the bits of a different parameter.

Ordinary native builds consume the committed SPIR-V artifacts after checking the exact shader-source and included-GLSL hashes, artifact identities, division-workaround variants, generated variant header and compiled kernel manifest. They do not silently regenerate numerical kernels with whichever shader optimizer happens to be installed on the build host. This matters across machines: optimizer versions can remove unused push-constant padding, and older SPIR-V tools may not understand an extension present in a retained kernel. The asset check rejects stale inputs; the separate rebuild/validation workflow requires a compatible toolchain and reviewed artifacts. Identity verification preserves the selected implementation; it does not replace device admission, dispatch parity or scientific validation.

The dispatch graph records read-after-write, write-after-read and write-after-write dependencies between named field uses. It is a hazard description, not an executable resource binder. A structurally valid graph still needs the correct buffers, extents, constant block, kernel and backend synchronization. This distinction keeps a graph validation pass from being misreported as successful GPU execution.

Memory admission is also backend-resolved. The largest individual field must fit the device's storage-buffer limit, while simultaneous resident fields must fit an available heap budget. Those are independent constraints. The current dynamics envelope uses 51 field-equivalents per simultaneously resident level, including retained names and transients; it is a conservative planning envelope, not the exact number of prognostic variables. Specialized wider transactions retain their own limits. Saturating arithmetic reports overflow as invalid admission rather than wrapping a huge request into a small one. See memory admission and its tests.

If $B_f = 4n_x n_y n_z$ is one field payload, the simple envelope is approximately $51B_f$ per level before interpreting host/device duplication. A hierarchy must sum all levels that can be resident together. A device-local heap size is not necessarily its currently free budget; Nimbus distinguishes a known heap from a live budget when the backend exposes one. Admission can be conservative, but it must not invent VRAM capacity from a CPU's system-memory report.

16 Vulkan transfers, execution and failure containment

16.1 Choose memory from reported types, not a GPU name

The Vulkan memory plan inspects compatible memory types and their heaps. It identifies a primary device-local heap, a working allocation type and a host-visible staging type. For staging it prefers cached host memory, which matters for CPU readback. Host coherence and host caching are different flags: coherence concerns visibility operations, while caching affects CPU access behavior.

A type that is both host-visible and device-local can describe genuinely shared memory, or a CPU-visible aperture into discrete VRAM. Resizable BAR does not turn the latter into an integrated-

memory machine. Nimbus keeps a discrete device in an aperture/staged classification even when the visible aperture spans the full device heap. Direct shared mode requires compatible reported topology; `auto`, `direct` and `staged` controls select among supported plans rather than bypassing their requirements. The underlying distinction is described in the Khronos memory-allocation guide.

The buffer pool retains capacity for repeated dispatches. In direct shared mode a device buffer is mapped and the staging slot can be a placeholder. In staged mode the pair contains host-accessible staging and device-local storage. Batch allocation can share a backing allocation while retaining separate buffer handles. Descriptor updates still bind each logical resource with its correct range. Retaining capacity reduces allocator work; it does not remove the obligation to upload newly changed data.

16.2 The staged sequence has two kinds of copies

A representative synchronous operation follows this sequence. It is a description of ownership stages, not pseudocode authorizing arbitrary barrier removal:

```
validate dispatch and obtain owned candidate buffers
copy host input arrays into mapped staging
flush noncoherent host writes when required
record staging -> device copies
barrier: transfer writes become shader-readable
record compute dispatches and inter-kernel dependencies
barrier: shader writes become transfer-readable
record device -> staging copies
submit command buffer; wait for confirmed completion
invalidate noncoherent host mappings when required
validate result; copy/publish accepted host outputs
```

The CPU copy into staging and the GPU transfer into VRAM are separate work. Similarly, downloading to staging and copying into the solver's authoritative host arrays are separate work. Telemetry distinguishes host upload/download bytes from device upload/download bytes. Adding those quantities without stating the boundary can double-count one logical field movement, while omitting either can hide the actual cost paid by the process.

A pipeline barrier describes ordering and memory visibility between producer and consumer accesses. A fence lets the host learn that submitted work has completed. A mapped-memory flush makes noncoherent host writes available; an invalidate makes completed device writes visible to a noncoherent host mapping. None is a synonym for the other. Host-coherent memory removes particular cache maintenance calls, not the need to wait before reading a result or overwriting in-flight input. Khronos's synchronization examples illustrate these distinct dependencies.

Nimbus aligns noncoherent ranges to the device's atom-size rules and avoids processing one shared allocation repeatedly through multiple handles. The authoritative implementation is the Vulkan backend. A change that removes a copy must also update the relevant authority and visibility transitions; deleting `memcpy` from one branch is not a complete residency optimization.

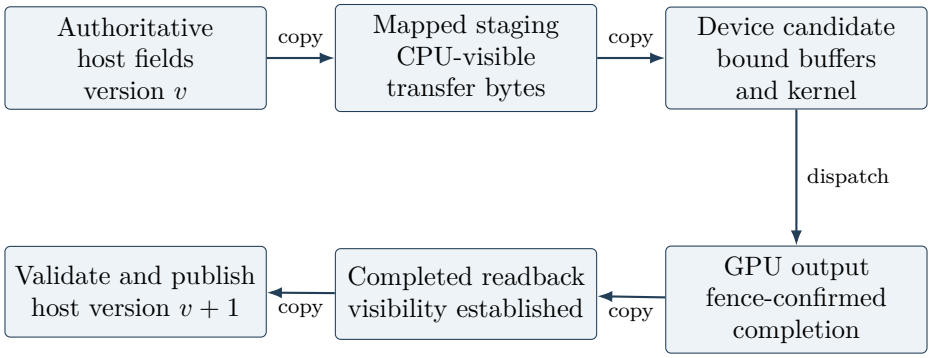


Figure 2: A representative staged-memory compute transaction. Direct shared memory can remove the separate staging transfer, but still requires ordered access and an authoritative version transition. The placement of validation and commit is an operator contract; uncertain GPU execution cannot release the resources it may still reference.

16.3 An uncertain submission cannot release its buffers

The current compute owner records and submits one shared command buffer at a time. A successful queue submission means the queue accepted work, not that its outputs are ready. Completed-operation counters advance after fence confirmation and required visibility handling. Kernel counts are associated with that completed submission, so a failed or abandoned command is not reported as successful native science.

If submission or waiting leaves execution uncertain, the backend is poisoned and referenced resources remain pinned until shutdown. Reusing a pool slot after a timeout could let the GPU write into a later unrelated operation. The pure submission lifecycle models this independently of real handles so fault tests can cover the decision. A host fallback is possible only where the enclosing runtime allows it and retains a separate valid host state; native-required admission does not become optional because a device failed.

17 Residency is a versioned authority graph

A field allocation can exist without containing the current field. Nimbus's residency graph records a logical version, host version and device version together with host identity and extent. A host write advances the logical version and host version; a device write advances the logical version and device version. Synchronization copies a known version to the other side. Reuse is valid only when the side's version equals the authoritative version and identity/extent still match.

For example, suppose both copies hold version 7. A host boundary update makes the host version 8 while the device remains at 7. A subsequent kernel cannot skip upload merely because its input pointer and allocation handle are unchanged. Conversely, after a device-only update produces version 9, a host diagnostic must not read version 8 just because the host pointer remains valid. The state machine exposes these mismatches; it does not itself execute a fence or inspect the pointed-to bytes.

ResidentFieldStore combines that authority graph with named allocation claims. Replacing allocator callbacks while claims are live is rejected. Batch acquisition publishes no partial set if a candidate allocation fails, and the callbacks must outlive the claims that use them. The store is not thread-safe: the backend must serialize its transitions. Its destructor intentionally does not call an

allocator that may already have been destroyed; owners needing individual release use the explicit clear path while the allocator is alive.

Generation identity is especially important for restart, grid movement, resizing and nested contexts. Pointer reuse by a host allocator is not proof that an old device value still represents the same simulation field. The residency tests cover state and allocation behavior, while full runtime tests must additionally prove that every actual writer announces its transition. A correct cache state machine cannot compensate for a producer that silently mutates storage behind it.

18 Metal follows the same ownership questions with different APIs

Metal exposes a device, command queue, buffers, pipeline states and command encoders rather than Vulkan's object model. Nimbus stores their session identity in MetalDeviceContext. Its synchronous finish path commits the command buffer, waits for completion, checks the final status and only then advances completion telemetry. GPU start and end timestamps describe the completed command; host submit/wait time is a different interval.

Shared-storage buffers make memory CPU- and GPU-accessible on the applicable device, but a host vector and a Metal buffer created with `newBufferWithBytes` are still different owned objects. Creating that buffer copies bytes. Copying its contents into a solver field after completion is also real work. Some operators retain resident named resources; others still allocate or copy through their ordinary paths. A blanket statement that Metal avoids all transfers would obscure the actual implementation.

Apple's storage-mode guide distinguishes CPU-visible shared resources from GPU-private resources. A private resource may require explicit transfer access when a CPU needs its data, while shared access still needs correct execution ordering. A private storage experiment does not become the production policy merely because a local benchmark can create such resources. Nimbus's advection owner and resident state show concrete copy and reuse paths.

The backend also checks kernel coverage against the canonical manifest. A kernel demanding native Float64 arithmetic is reported as hardware-limited in the current Metal coverage contract; it is not secretly replaced by a Float32 kernel and called equivalent. Other scientific kernels use explicitly maintained compensated Float32 formulations. Availability of one family says nothing about an unrelated family's numerical requirements.

19 Vulkan and CUDA: existing assistance and a future hybrid

Nimbus can explore NVIDIA-specific computation without making NVIDIA hardware a prerequisite for its portable solver. There are two different technical routes: launching a limited PTX kernel from a Vulkan command buffer, which the source already implements for retained acoustic flux, and sharing explicit resources between Vulkan and the CUDA runtime, which would require additional engineering. Neither route establishes an arbitrary CUDA solver backend or multi-GPU domain decomposition.

Performance work should examine a specific operation and the workload around it. CUDA may reduce the work required, provide a better implementation or make a suitable library available while preserving Nimbus's numerical and ownership contracts. Crossing an API boundary also adds dispatch and synchronization costs. Complete-step measurements must establish whether the

combined design improves the actual simulation.

19.1 The narrow path already present in Nimbus

The strict-v1 selector `extensions.compute.cuda_assist` defaults to false and requires the Vulkan backend. It does not add `cuda` to the supported backend names. Its kernel owner loads the retained-flux PTX module, obtains the function entry point and records its dispatch inside a Vulkan command buffer using `vkCmdCudaLaunchKernelNV`. The strict retained-flux path uses Vulkan buffer device addresses for its density, wind and flux buffers. The rest of the solver's scientific profile and native ownership requirements remain in force.

Initialization checks the selected NVIDIA device, Vulkan extension availability, buffer-device-address support and the CUDA-launch feature. The current helper requires compute capability 7.5 or newer. Missing header/API support, unavailable entry points or module creation failure rejects that backend initialization. The general runtime can disclose a CPU recovery when the profile permits `extensions.compute.allow_fallback`; native-required execution must have effective fallback disabled. Merely requesting assist is therefore not proof of its execution. The active path logs the retained-flux PTX digest, and its manifest identity includes that digest.

The readiness registry names this limited retained-flux capability separately. Its source and bounded evidence are not a statement that every operator can run through CUDA, that a long hardware campaign has passed, or that a broader hybrid has completed scientific review. The Khronos extension description explains that this mechanism dispatches PTX through Vulkan without creating the usual Vulkan/CUDA context-interoperability path. It is a provisional, NVIDIA-specific extension, so driver and header availability remain part of admission and release maintenance.

19.2 A different future route: shared external resources

A more general hybrid could keep Vulkan as the owner of ordinary solver dispatch and use a CUDA runtime stream for a selected operation or library. The proposed first scope is one process, one physical NVIDIA GPU and explicitly shared buffers. It is not implemented by the existing assist flag. A future implementation would need a distinct, opt-in public capability and an exact supported profile before users could select it as a normal execution path.

NVIDIA's Vulkan interoperability guide describes importing exported memory and synchronization handles into CUDA on the same physical device, identified by matching device UUIDs. An ordinal such as device zero is insufficient because the two APIs can enumerate devices differently. On a laptop, Studio's webview may render on another adapter from the solver. This proposed sharing concerns solver allocations; it does not give the separate Studio process direct access to them.

The aim would be to avoid a full device-to-host-to-device round trip around the selected operation. The shared allocation remains GPU-accessible while the two APIs exchange permission to use its current contents. "Shared" does not mean that both may write simultaneously or that a completed host API call proves the device work finished. A producer/consumer protocol is still necessary.

19.3 Admission must query the actual allocation contract

A proposed interop session would first identify the same physical device in both APIs and check the supported operating-system handle types. It would then query the intended resource, not just the device's marketing identity. Khronos's external-buffer query includes the buffer's usage, creation

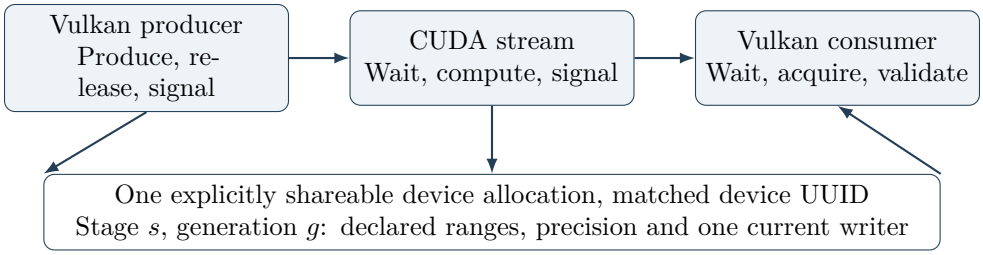


Figure 3: Proposed CUDA-runtime interoperability on one physical GPU. External memory avoids a host round trip; external synchronization and ownership still order access. This is distinct from Nimbus's existing Vulkan-command-buffer PTX retained-flux path and from future multi-GPU domain decomposition.

flags and external handle type. The returned memory properties describe export/import features and compatible handle types. A valid design must honor dedicated-allocation requirements as well as ordinary alignment and memory-type requirements.

For Nimbus, the imported object's record would need allocation size, field byte range, element precision, strides, alignment, domain generation and owner. A suballocation offset cannot be dropped simply because two APIs refer to the same underlying memory. The CUDA import's dedicated-resource flag must agree with how Vulkan actually allocated the resource. CUDA buffer mappings and image mappings are different interfaces: a tiled image cannot be assumed to have a linear row layout merely because its logical shape matches a field. An initial buffer-only scope would reduce the number of layout assumptions that need to be proved.

The CUDA external-resource API reference specifies mapping offsets and alignment, separate release of mapped buffers, and handle ownership. Successful opaque-FD imports transfer descriptor ownership to CUDA; ordinary Win32 NT handles retain application-managed handle lifetime. Outstanding semaphore operations must finish before their objects are destroyed. Nimbus would need separate unwind paths for failure before import, after import, after mapping and after submitted work, so an error does not leak an allocation or close a handle twice.

An external-memory handle also does not make host access automatically coherent. If the selected policy maps the allocation on the CPU, the required noncoherent flush/invalidate operations and barriers still apply. The runtime API documents this requirement for CPU-mapped Vulkan memory. A host diagnostic or output writer needs a published, completed endpoint, just as it does in the existing Vulkan residency design.

19.4 Ordering two APIs without serializing the whole device

The proposed operation would have an explicit dependency chain: Vulkan produces the input range and releases the required ownership; an exported semaphore signals that endpoint; the CUDA stream waits, executes the selected operation and signals completion; Vulkan waits and reacquires the resource before its next consumer. The exact barriers, stage/access scopes and any image layouts must follow the resource type and external-ownership rules. The Vulkan synchronization specification distinguishes execution order, memory visibility and queue-family ownership. One of those requirements cannot stand in for all the others. Where the sharing contract requires external ownership transfer, the barrier uses the applicable external queue-family identity, such as `VK_QUEUE_FAMILY_EXTERNAL`; a CUDA device ordinal is not a Vulkan queue-family index. Image layout and ownership must be tracked separately.

The external-semaphore capability query must admit the selected semaphore and handle type. Binary semaphores need a correct signal/wait reuse protocol; a timeline design additionally needs support for its exported handle and a consistent monotonically advancing value scheme. The CUDA guide describes both cases and their cross-API signal/wait ordering. For Nimbus, a timeline value would be associated with a concrete field generation and integrator stage rather than treated as an unexplained global counter.

CUDA's asynchronous external-semaphore signal is ordered in its specified stream after earlier work in that stream. A library operation on another stream is not thereby complete. The hybrid owner would set the library's stream explicitly or insert the required cross-stream dependency before signaling. Similarly, submitting a Vulkan command buffer does not establish that the CUDA wait has completed. Full-device idle waits can provide a simple diagnostic baseline, but using them around every operation could erase the benefit of interoperability. Any later overlap optimization must retain the same dependency graph and failure semantics.

19.5 Which computations might justify the additional path?

Candidate work should be chosen from measured solver costs. A substantial well-defined numerical operation with stable input/output buffers is a better initial experiment than repeatedly switching APIs around tiny pointwise kernels. A batched operation can amortize launch and handoff overhead; a long sequence of small transfers can instead increase latency and duplicate scratch storage.

NVIDIA provides cuBLAS for dense linear algebra, cuSPARSE for sparse operations and cuFFT for Fourier transforms. These are possible building blocks, not evidence that Nimbus currently needs or calls them. Mapping a stencil, column solve or diagnostic to a library routine is worthwhile only when its formulation, boundary conditions, precision and data-layout costs fit the actual algorithm. Introducing a spectral method, for example, changes more than the dispatch API if the original operation was not mathematically spectral.

CUDA can also provide a useful development environment for a specialized kernel. That does not automatically make its generated arithmetic faster on the target GPU, nor remove the need to maintain the portable Vulkan, Metal and CPU implementations. The maintenance cost includes another compiler/runtime matrix, library versions, failure paths, scientific parity and hardware tests. A future optional path should leave builds without the CUDA toolkit functional.

19.6 Memory topology and multiple devices

The benefit of avoiding host round trips can be especially relevant when a discrete GPU has its own memory and a finite host interconnect. On a unified system, copies and synchronization still have measurable costs, and the two API contexts do not automatically share every allocation. Unified CPU/GPU memory, CUDA managed memory and explicitly exported Vulkan memory are distinct mechanisms. This NVIDIA-specific proposal does not establish CUDA support on Apple GPUs or turn ARM64 into a proxy for unified-memory behavior.

Same-device resource sharing also does not establish peer access. Two installed GPUs need their own device identities, import/transfer capabilities, synchronization and ownership. An allocation imported for one device is not a promise that another device can use it. The multi-GPU proposal adds physical-domain partitioning, halo exchange, reductions and continuation ownership beyond this single-device experiment. A CUDA-aware communication library would require evidence for the actual imported allocation and transport; its name alone does not validate arbitrary Vulkan buffers.

19.7 Scientific and failure-atomicity requirements

A future hybrid operation would consume the same stage, carrier, physical domain, boundary state and private moments as the existing owner. It would produce the same declared outputs and accounting. Duplicating only the main visible field while leaving a private reservoir or diagnostic on an old endpoint would not be an implementation of the original operation.

Precision must be explicit across the boundary. NVIDIA's floating-point white paper explains why fused operations, intermediate rounding and reduction order can change results even with IEEE arithmetic. Nimbus would retain its declared storage and intermediate types, FMA contract, denormal behavior and reduction requirements. Library math modes or reduced-precision accelerators could not be enabled merely to obtain a faster timing. Any intentional numerical change would need its own scientific justification and tests.

Tests would include direct references, limiting and positivity, conservation or closure, invalid inputs, threading behavior and CPU/Vulkan/Metal parity where applicable to the declared profile. They would also exercise an error after Vulkan production but before CUDA completion, and an error before Vulkan reacquisition. A rejected transaction must leave the previous committed state, private fields, reports and generation ownership consistent. Restart must capture the full continuation endpoint after both APIs finish, not a snapshot taken between them.

The public configuration, catalog, readiness record and runtime dispatch report would need to identify the composite path and its exact supported scope. Dispatch evidence should name the selected operation, device UUID, kernel or library identity, precision, shared/staged transfer path and synchronization outcome. Unsupported imports or capabilities should reject an explicitly required hybrid profile. An intentionally permitted fallback must be disclosed; it cannot count as a successful native hybrid experiment.

19.8 An experiment needs a complete cost model

For a proposed operation, an illustrative break-even condition is

$$T_{\text{Vulkan operation}} > T_{\text{CUDA operation}} + T_{\text{handoff}} + T_{\text{layout}} + T_{\text{extra synchronization}}. \tag{1}$$

The right-hand terms include real preparation and ownership work. Persistent mapping can amortize setup over many steps, but its retained memory and teardown still count. A steady-state measurement that excludes compilation or import must report those startup costs separately.

The bounded performance gate should compare the same source/configuration, scientific endpoint, grid and output cadence on the same hardware. Record compiler flags, driver and library versions, clock/power conditions, CPU worker settings, memory topology, warmup and repeated-trial variation. Measure the selected operation, complete solver step, host synchronization, output/restart cost and peak memory. A faster kernel that moves the bottleneck to publication or increases memory pressure can lose end to end.

Only after ordinary public selection, admission, engineering and bounded tests are complete would this new capability be Ready for its required long campaigns. Campaign validation and independent scientific review remain separate steps toward Fully Qualified. The present design analysis supplies a path to test, not a speedup promise or a declaration that external-resource interop is shipped.

20 Sharding a simulation across multiple GPUs

This section is a design analysis of work Nimbus would need. It does not describe an implemented distributed solver. The current backend configuration selects one device index for a compute runtime. The Vulkan context owns one selected physical device, logical device and compute queue; the Metal context owns one device session and command queue. There is no maintained MPI domain decomposition or complete multi-device timestep owner in the ordinary runtime. Multiple runtimes, nested domains, CUDA-assisted kernels and remote SSH execution do not, individually or together, establish that capability.

20.1 What is being divided?

There are three substantially different designs. Independent simulations can run on different devices without exchanging atmospheric state. Spatial decomposition instead divides one atmosphere into owned subdomains, which must exchange the state needed by neighboring numerical operators. Assigning different physics packages to different GPUs is a third design: for example, dynamics on one device and radiation on another. That arrangement must move inputs and return tendencies at their declared cadence. It may duplicate much of the state and adds a coupling dependency, so it should not be assumed to solve a memory-capacity problem.

For an initial spatial implementation, horizontal tiles with complete vertical columns are a reasonable candidate. They keep column radiation, boundary-layer solves, vertical sedimentation and a column-local implicit solve on one owner. This is a proposed starting restriction, not proof of support for every current column formulation. It can produce poor aspect ratios or leave too few tiles for a large machine. Splitting vertically later would require explicit distributed column algorithms or gather/scatter phases, including their memory and synchronization costs. Terrain and moving nests make ownership more complex even when columns remain whole.

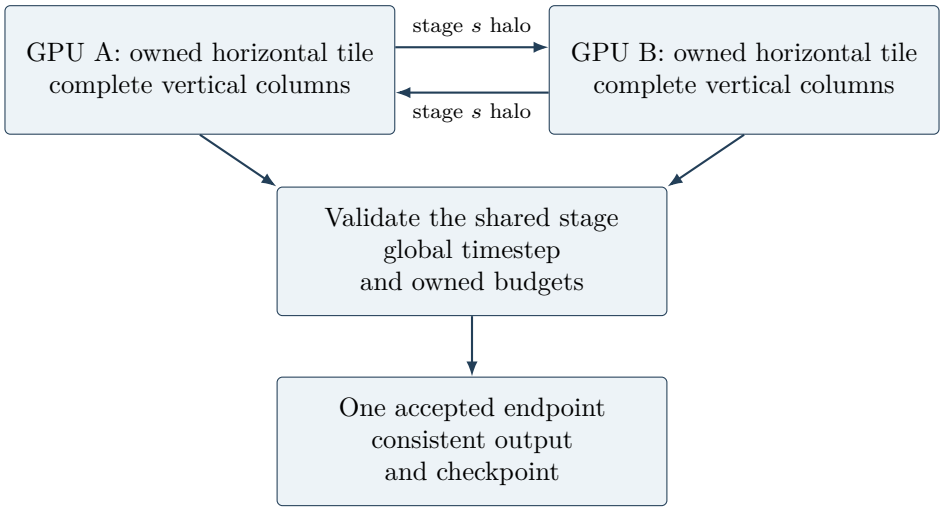


Figure 4: Proposed two-tile solver decomposition. Halo exchange can use a validated direct path or explicit host staging. Arrows describe data dependencies, not a mandatory global barrier after every local operator. Each tile owns its physical cells once; copied halos carry no additional atmospheric inventory. This is a design diagram, not a current Nimbus feature.

20.2 Own physical cells and faces before assigning buffers

Each tile needs global physical bounds, local storage bounds, halo widths, neighbors and a stable mapping from global to local indices. A halo is a readable copy of another tile's state, not additional atmospheric volume. Inventory reductions must count owned cells once. On a C-grid, a velocity or mass-flux face shared by two tiles also needs a canonical owner or an equivalent reconciliation rule. Independently evaluating slightly different fluxes on both sides can create an interface source even when each local kernel passes its own conservation test.

For one mass flux Φ_f oriented from tile A to tile B, conservative exchange requires

$$\Delta M_A = -\Delta t\, \Phi_f, \qquad \Delta M_B = +\Delta t\, \Phi_f.$$

The same face, time interval, stage weight and transported carrier must be used on both sides. Water mixing ratios, number per dry mass and number per volume cannot share an unspecified scalar convention. Geometry, metric coefficients, held forcing and scheme-private fields require exchange wherever their consumers need them. A fixed list of visible plot fields is not a sufficient halo protocol.

Messages should identify the grid or hierarchy generation, physical interval, integrator stage, field and location, source and destination ownership, extent and precision. A completed transfer with the wrong stage is stale data. Halo width must follow the largest actual dependency footprint of the selected operator, including reconstruction and corner dependencies; it cannot be guessed from the nearest-neighbor appearance of the final divergence.

Physical boundary conditions apply at the atmosphere's external boundary. An internal partition face must not become a wall, sponge or independent open boundary. Periodic neighbors must close the global topology. A processor partition on a fixed grid is also different from a coarse/fine nest: the latter needs metric-aware restriction, prolongation and flux reconciliation, not merely a copy between equal-resolution halos.

20.3 Exchange at the stage that consumes the data

A portable first implementation could establish this sequence for each operation with neighbor dependencies: complete the producing work, pack the required owned boundary state, start its transfers, compute independent interior cells, complete and unpack the receives, then compute the boundary region. The new endpoint becomes visible only after all required regions and validations complete. Packing, transfer and unpacking each retain buffer ownership until their respective completion events.

This sequence must be derived from the actual split integrator. Acoustic substeps may require pressure and face-wind exchange several times during one slow step. Runge–Kutta stages consume their own stage state. Exchanging only after a complete outer timestep is generally insufficient. A global CFL decision must use the most restrictive participating tile; a local reduction does not establish the safety of the global timestep. Iterative diagnostics or solvers can add global reductions beyond those required by local stencils.

The current dispatch graph describes field hazards, but is not a distributed execution engine. It would need region and device ownership, explicit exchange edges and completion dependencies. Likewise, the residency contract would need to distinguish an owned interior version from a halo version for a particular stage. Marking an entire array "current on the device" cannot express that one boundary slab has not arrived.

20.4 Portable communication first, optional direct paths second

The reference transport should work with explicit device-to-host transfer, owned host send/receive buffers and host-to-device transfer. This gives a testable route without assuming peer access. On a single machine the exchange can be managed within one process or across explicitly owned processes. Across machines, MPI is a plausible transport and collective layer. Its nonblocking neighborhood operations provide an interface for neighbor exchange, but posting a nonblocking operation is not proof of overlap or completion. Progress, thread support and the application's wait schedule must be measured and validated. The transport does not know Nimbus's field units or stage semantics.

Direct device transfers are optional optimizations of the same exchange contract. Vulkan device groups are enumerated compatibility groups; they do not promise that arbitrary installed GPUs can be treated as one memory pool. The Khronos device-group guide and peer-memory query describe the relevant capability checks. Separate logical devices instead require a supported sharing/transfer mechanism with explicit lifetime and synchronization rules. A queue barrier or fence on one independent device does not by itself establish that another device can consume the bytes.

Metal similarly requires using the devices actually exposed by the system. Apple documents multi-GPU systems and transfers between connected GPUs. Availability depends on hardware and API support; multiple packages or dies inside a product do not imply multiple independently selectable Metal devices. Unified CPU/GPU memory on one device does not establish peer access to another computer or accelerator.

"GPU-aware MPI" also needs a specific memory API, transport and build. For example, Open MPI's CUDA support describes supported CUDA-buffer paths. It does not certify arbitrary Vulkan allocations or Metal buffers. Nimbus would need a validated interoperability path or host staging for those owners. An unavailable direct path may select an explicitly supported staged transport; it must not silently change the numerical operation or violate a native-execution requirement.

20.5 Why more memory does not imply linear speedup

Let an illustrative cubic owned tile have side n, halo depth h, F communicated scalar-equivalents and b bytes per scalar. Its owned payload is bFn^3. A complete surrounding halo occupies

$$B_{\text{halo}} = bF\big[(n+2h)^3 - n^3\big] = bF(6hn^2 + 12h^2n + 8h^3).$$

If only the six face slabs are communicated in an exchange, their leading payload is $6bFhn^2$; corners, staged directional exchanges and non-scalar locations change the exact count. This is a teaching model for an interior tile, not the traffic of a particular Nimbus kernel. As a fixed global problem is divided into more pieces, the surface-to-volume ratio grows. Full-column horizontal tiles have a different geometric count but the same concern.

For message startup cost α, effective transfer bandwidth β, m messages and payload B, a simple model is

$$T_{\text{exchange}} \simeq m\alpha + B/\beta + T_{\text{pack}} + T_{\text{unpack}}.$$

With perfect interior/exchange overlap, and with boundary work and a collective following those phases, a stage still takes roughly

$$T_{\text{stage}} \gtrsim \max(T_{\text{interior}}, T_{\text{exchange}}) + T_{\text{boundary}} + T_{\text{collective}}.$$

Real overlap can be worse: packing can contend for memory bandwidth, transfer engines may share links, and small tiles can underfill a GPU. Frequent acoustic exchanges can become latency-bound.

A second board behind a narrow or shared link may increase capacity while adding little useful throughput.

Every device must fit its own owned state, halos, stages, rollback and communication buffers. The sum of nominal VRAM capacities is not an admission test. Nor should every rank retain a full global host mirror: that would preserve a memory bottleneck even after the device fields were partitioned. NUMA placement should associate host staging buffers and communication threads with the relevant device and network path. On heterogeneous GPUs, equal cell counts need not give equal stage times. Cloud-dependent work and nested placement can also make the slowest tile change during an experiment. Dynamic balancing would require a state migration transaction and new generations, not just moving an array pointer.

20.6 Global diagnostics, restart and failure must agree

Budget reductions should accumulate owned physical inventories with declared precision and ordering. A reduction tree may change rounding when the number of ranks changes. Fixed-decomposition repeatability, decomposition-independent bitwise equality and bounded scientific agreement are distinct claims. They need distinct tests; ordinary parallel summation does not promise all three. Random perturbations should have a defined mapping to physical identities and time if results are intended to be independent of rank placement.

A checkpoint needs a globally consistent accepted endpoint: all owned primary and private fields, geometry, held-source clocks, histories, configuration and publication identity. Repartitioned restart needs a mapping from saved physical ownership to the new layout; it cannot depend on finding the same GPU index. Partitioned output should publish a manifest only after all required pieces are durable, and readers should reconstruct the logical field without treating rank order as a coordinate system. Gathering the whole atmosphere on one rank for every frame can become a new memory and I/O bottleneck. Existing NPY, netCDF and ZFP support does not by itself supply this distributed publication protocol, and this design does not authorize removal of NPY.

If one tile rejects a timestep, neighboring tiles cannot publish a mixture of old and new endpoints. A first design should stop the distributed job on an unrecoverable device or communication failure and resume from a validated global checkpoint. Transparent continuation after losing a rank is additional engineering, not a consequence of using MPI. Buffers involved in uncertain device submissions must remain owned until completion is known or the corresponding execution context has been safely retired.

20.7 A sequence that can be implemented and reviewed

Begin with the distributed state and exchange contract, including a one-tile path that reproduces the existing admitted experiment. A two-tile CPU or host-staged reference can then expose ownership errors independently of peer transport. Manufactured fields should cross each partition orientation and corner; interface inventories must cancel, and halos must never enter physical budgets. Tests should mutate stage identities, interrupt transfers and reject partial publication.

The next bounded target is a two-device, fixed-grid configuration with stated physics and boundary conditions, selected through ordinary public configuration. It needs whole-step one-device comparisons, native dispatch evidence, restart on the same decomposition, rejected-input atomicity and memory accounting. Changing the decomposition on restart, nonuniform hardware, terrain, private microphysics state, held radiation, nesting and moving domains each extend the contract and require their own completed coupling. They cannot be called supported merely because the flat

dry exchange works.

Only after correctness should direct peer transfers, overlap, larger device counts and multi-node transports be assessed with strong- and weak-scaling experiments. Strong scaling holds the physical numerical problem fixed; weak scaling holds approximately the work per participant fixed and generally grows the global problem. Report both completion time and resource cost, including output and restart. Source code or a two-GPU speedup alone is not Ready, and Ready remains separate from campaign validation and independent scientific review. This staged design keeps the ordinary local workflow usable while making distributed execution an explicit, testable extension.

21 Floating-point semantics across CPU and GPU

21.1 Storage precision, arithmetic precision and rounding points

Binary32 has 24 bits of significand precision, including its implicit leading bit for normal values; binary64 has 53. Their unit roundoffs near one are approximately 2^{-24} and 2^{-53} under round-to-nearest. Storing the main fields as Float32 therefore limits what a wider intermediate can publish. If a source computes a tiny increment in double precision and then stores `float(old + increment)`, that increment can disappear at the storage boundary. It is not necessarily safe to move the cast earlier, because an intermediate rounding can also change a threshold decision or cancellation.

Nimbus's dry EOS deliberately evaluates from stored Float32 density and theta using wider arithmetic before storing pressure. Energy attribution distinguishes ideal changes from water, theta and pressure storage effects. Backend work must preserve the intended rounding points and constants. A tolerance patch that makes one Linux failure disappear without explaining the changed operation would conceal a numerical portability problem rather than solve it.

21.2 FMA is useful precisely because it changes rounding

A fused multiply-add evaluates $ab+c$ with one final rounding, whereas separate operations compute a rounded product before the sum. Schematically,

$$\mathrm{fma}(a,b,c) = \mathrm{RN}(ab+c), \qquad \mathrm{RN}(\mathrm{RN}(ab)+c) \text{ need not be equal.}$$

The distinction is especially visible when ab and c nearly cancel. Compensated algorithms can intentionally use FMA to recover a product residual: if $p = \mathrm{RN}(ab)$, then $e = \mathrm{fma}(a,b,-p)$ captures the remaining portion under the algorithm's finite-range assumptions. Reassociating that expression or replacing it with an ordinary multiply and subtract defeats the construction.

This requirement extends into the shader binary. The GLSL extended-instruction specification permits different evaluation of an FMA without the corresponding `NoContraction` constraint. Nimbus's canonical shader build therefore preserves that constraint on explicit FMA instructions, validates the resulting module and binds its exact artifact identity to the runtime catalog. Checking the source spelling alone would miss this distinction. Ordinary Float32 FMA constraints and the stricter device-gated FMA extension are separate mechanisms.

Even a helper named `fma_vectors` needs its concrete implementation examined. Nimbus uses the fused AVX intrinsic when `__FMA__` is compiled, separate multiply/add in the SSE and non-FMA AVX branches, and a Neon fused intrinsic in its AArch64 vector branch. Scalar tails have their own compiler contraction behavior. Thus a name alone does not establish bitwise-identical evaluation on every architecture. Scientific kernels with stronger requirements use their own explicit arithmetic contracts.

21.3 Compensated pairs and device capability gates

Several GPU kernels represent a more accurate intermediate as a high/low Float32 pair. Addition and multiplication propagate low-order residuals with carefully ordered fused operations. This increases useful precision for a defined calculation, but it is not a universal replacement for IEEE binary64: the exponent range, subnormal behavior, transcendental functions and final storage conversion still need their own treatment. The mapped scalar Metal helper is generated from shared arithmetic sources rather than being separately hand-tuned into a different equation.

Evaluation order and operation accuracy are separate requirements. Vulkan's floating-point accuracy rules do not give ordinary GLSL FMA the correctly rounded IEEE guarantee that a compensated product residual would need. A driver can return zero for that residual even when the exact product has a nonzero low part. Nimbus's portable product compensation therefore splits the operands into high and low significand pieces and preserves the required ordering of the partial products. This avoids relying on a stronger FMA accuracy promise than the selected instruction supplies. Overflow, underflow and the finite precision of the two stored components still define the usable arithmetic range.

Subnormal handling needs a separate safeguard. A Vulkan device may flush a subnormal input or product to zero even when its ordinary normal multiplication is correctly rounded. The compensated product helper detects those cases from the Float32 exponent bits. It multiplies the two 24-bit significands in 12-bit pieces, retains the exact 48-bit integer product, and rounds the result to Float32 with ties to even. Normal products keep the ordinary multiplication path. This small integer fallback needs neither Float64 nor 64-bit integer shader support, and preserves signed zero and nonfinite classification. The product and its residual are checked separately against CPU arithmetic; this does not grant the entire compensated library a universal IEEE guarantee. The software product path is enabled for the acoustic pressure and momentum calculations and Earth-control correlated noise. Other physics kernels retain their existing compensated arithmetic, including capability-gated native FMA variants where available. Extending software compensation to those kernels requires their scientific error checks and compiler-memory and throughput validation; the small primitive probe does not establish those larger contracts.

Compiler working memory is also part of portability. Repeatedly inlining the compensated routines and expanding their series loops can turn a small source kernel into a very large compiler input. For the portable-product kernels, the canonical SPIR-V build retains helper functions, marks them `DontInline`, and marks their series loops `DontUnroll`. These are compiler controls, not changes to the numerical series, its order or its precision. Native parity and compilation checks must still run on each target because a driver's treatment of optimization hints is implementation dependent. Other kernels retain ordinary function-inlining choices: preserving every call boundary can itself increase a driver's compilation cost. A compact shader module alone is not a throughput qualification.

Boundary reconstruction has a related ordering requirement. A ghost value is formed as its local base state plus the interior perturbation, $b_g + (x_i - b_i)$. Computing $(b_g + x_i) - b_i$ can lose a bit even when the interior perturbation is zero. The native A-grid boundary paths preserve the former grouping, and a balanced-column regression checks that a zero perturbation leaves the base profile unchanged.

Vulkan distinguishes ordinary Float32 kernels from those explicitly requiring Float32 storage with Float64 arithmetic. The latter are gated on the required Float64 rounding/preservation properties, strict Float64 FMA and integer storage-conversion capability. Compensated families also require the relevant Float32 signed-zero/NaN preservation and round-to-nearest controls. A device adver-

tising basic compute support or a high peak throughput number does not thereby satisfy these scientific requirements.

Tiny values introduce another issue: devices can flush subnormal arithmetic to zero. The maintained storage helper scales tiny inventories before arithmetic and uses explicit integer-bit rounding for the subnormal storage case. That is an algorithm-specific representation technique with tests, not permission to add platform-specific epsilon clamps to every failed comparison. Metal's retained compile policy disables fast math for these owned shader sources; generator checks keep translated arithmetic synchronized.

The practical validation sequence follows the contract: check invalid inputs and output atomicity, compare the intended operator against its independent reference, test native and generic CPU builds where applicable, then exercise the real selected backend and memory policy. Exact saved-state comparison, bounded numerical tolerance and scientific closure remain distinct evidence. Neither a portable build nor a parity test alone proves that every tracked configuration is Ready.

22 Publication, compression and persistent state

22.1 A frame is published by a protocol

The output layer translates a completed model state into selected products. It resolves the field inventory and units, stages required values, encodes their storage representation and publishes metadata identifying the completed frame. The existence of a directory or one field file is not enough to tell a reader that all required fields belong to one finished step. Studio's frame discovery uses publication state and manifest identity to avoid displaying a mixture of old and partially written new data.

AsyncOutputWriter has synchronous and asynchronous paths. A synchronous writer can use a valid borrowed view while the caller holds the state stable. A background writer requires owned snapshot storage; `submit` rejects a borrowed field rather than letting the integration thread mutate data while compression reads it. Its queue has a one-frame ownership bound, and the ordinary runtime waits for availability before assembling the next full snapshot. Backpressure is deliberate: an unbounded queue would convert a slow filesystem into uncontrolled RAM growth.

An asynchronous writer does not remove serialization or compression work. It allows some work to overlap while introducing a snapshot, queue synchronization and a separate error path. A write failure sets an error state and wakes waiting participants so that an output error cannot strand the producer indefinitely. The scientifically relevant frame remains identified by model time and source state, not by the wall-clock moment when the filesystem completes its write.

NPY provides a direct array format, netCDF provides structured scientific history, and ZFP provides selected compressed representations. Their wrappers must preserve axis order, shape, field identity and physical metadata. Internal storage order need not be the order a downstream VTK image expects; that conversion belongs to the reader/preparation boundary. A decoder that produces the right number of floats in the wrong order has not preserved a field.

22.2 What ZFP contributes, and what Nimbus adds

Peter Lindstrom's 2014 paper, *Fixed-Rate Compressed Floating-Point Arrays*, develops a block representation intended to make compressed numerical arrays accessible at a prescribed storage rate. Small 4^d blocks in d dimensions share a floating-point representation, undergo a decorrelating

transform and are encoded by significance. The paper supplies the codec method; it does not specify Nimbus's history container, keyframe schedule, field tolerances or publication transaction. See the primary paper.

The current library's modes answer different questions. Fixed rate constrains bits per value/block, while error depends on the data. Fixed precision controls encoded transform precision; it is not a uniform pointwise relative-error promise for every value in a mixed-magnitude block. Fixed accuracy targets an absolute error scale in the field's units and produces variable-length output. Reversible mode supplies bit-preserving reconstruction. These mode distinctions are defined by the versioned ZFP 1.0.1 mode reference. The library also documents finite-representation limits for exceptionally small accuracy tolerances. Zero tolerance in a lossy mode must not be described as equivalent to reversible compression; see FAQ questions 17, 20 and 21.

Nimbus calls the corresponding stream setters in its output writer and records the mode parameters in its own container. Ordinary public mode parsing accepts `accuracy`, `precision` and `rate`; admission imposes additional restrictions for a supported scientific profile. The retained ordinary admitted lossy-history contract requires accuracy mode and per-field tolerances, prohibits a positive sparse threshold and disallows the optional Float16 prefilter. Configuration code also caps the potential-vorticity tolerance at 10^{-8} in its SI representation: a generic visual tolerance could erase a meteorologically significant signal while still being a valid codec setting. These are Nimbus decisions in output configuration, not universal recommendations from the compression paper.

The optional sparse and Float16 transforms are separate lossy operations before compression. The former sets eligible small values to zero; the latter rounds eligible values to reduced significand precision. A codec tolerance applies to the values supplied to that codec. It does not retroactively bound information already removed by a prefilter. For a source x, prefilter F, codec reconstruction $D(C(F(x)))$ and later reconstruction R, a useful accounting separation is

$$|x - \hat{x}| \leq E_{\text{prefilter}} + E_{\text{codec}} + E_{\text{reconstruction}},$$

when the individual terms have actually been defined and bounded for that path. Writing down the inequality does not prove all three bounds. This is why the admitted policy restricts transforms and why quantitative output checks must compare the final decoded field, not just inspect a mode name.

Temporal delta history is another wrapper-level choice. The encoder keeps decoded previous fields so that the reader and writer predict from the same represented history. For a linear two-frame predictor, the conceptual equations are

$$\begin{aligned} P_n &= 2\hat{x}_{n-1} - \hat{x}_{n-2}, \\ r_n &= x_n - P_n, \\ \hat{x}_n &= P_n + \hat{r}_n. \end{aligned}$$

The implementation computes residual and reconstruction expressions with specified wider intermediates and Float32 storage. With identical predictor history, old error is already incorporated in the residual definition; using uncompressed previous fields at the writer and decoded previous fields at the reader would introduce a different drift mechanism. Actual error still includes residual rounding, codec error and final reconstruction rounding. Keyframes reset history dependencies, and retention cannot remove a frame that a retained delta chain still needs.

Nimbus's spatial container bricks are also different from ZFP's internal 4^d codec blocks. The current bricked reader can select $64 \times 64 \times 32$ container bricks, inspect a directory of offsets and reconstruct

only the required spatial region for supported versions. Legacy containers can require full-volume reconstruction. A small requested output window therefore does not always imply a small decode working set. The native reader reports these capabilities instead of silently labeling a full decode as a cheap window.

Field-parallel publication assigns independent fields to workers with isolated codec workspaces; libzfp execution within each field remains serial in this policy. It is not the same as enabling the library's internal parallel codec. Scratch budgets and exact serial/parallel artifact checks are part of the wrapper contract. More workers may improve throughput, increase memory pressure, or do neither; the retained benchmark harness makes no universal throughput claim. The roundtrip tests and benchmark contract tests address different parts of this behavior.

The reversible radar-v4 container uses the library's reversible mode and per-brick integrity information under a separate guarded mixed-history path. It must not be used to claim that ordinary lossy output is exact, or that the general radar-v4 readiness work is complete. Output history is also not the solver's in-memory representation: this architecture does not generally compress prognostic arrays in place between numerical operators. Paper results about compressed numerical-array computation cannot be transferred to Nimbus's history writer without a separate implementation and measurement.

22.3 Checkpoints encode continuation, not just visible fields

A restart needs more than wind, temperature and water arrays. It can require scheme-private particle state, time/cadence counters, boundary history, nested execution state, trajectories and other runtime owners. The checkpoint layer serializes a versioned set of sections rather than writing the raw bytes of a C++ object graph. Raw pointers, allocator state, vtable addresses and mutexes have no portable meaning in another process.

The wire format uses explicit little-endian integer encodings and IEEE floating-point values, section identifiers, versions, lengths, flags and checksums. It rejects duplicate or unsupported required sections and checks sizes before allocation. The architecture-specific in-memory layout is therefore separated from the continuation format. A native reader can convert byte order where needed; compatibility still depends on supported versions and the semantic state contract, not merely on recognizing the file magic.

The writer streams sections into a temporary file while computing integrity information, closes and synchronizes it, atomically publishes the destination, then synchronizes the containing directory through platform helpers. This avoids treating a partial file as the final checkpoint. On restore, decoded candidate state and attachments are validated before replacing the live state. An unknown future section or inconsistent tracking authority must not become a partially restored simulation. See checkpoint assembly and restore, wire tests and continuation tests.

23 Studio IPC, asynchronous work and remote execution

23.1 Keep native resources behind the Rust boundary

Studio's React components express user intent and maintain view state. Rust owns native filesystem and process operations. Tauri commands carry typed requests and responses across that boundary. A field-window response uses a compact binary envelope with a small validated header and Float32 payload, rather than inflating millions of values into a JSON number array. Geometry may add aligned Float64 components when required by the declared schema.

The native command checks source dimensions, range, stride, output count, arithmetic overflow and byte limits before serving the request. Its current window limits distinguish 64 MiB of returned field data from a larger bounded source working set. A shared native I/O reservation budgets source bytes plus twice the output bytes under a user-adjustable aggregate limit, then releases the claim through RAII when the last owner drops it. The automatic policy uses reported RAM; Settings exposes the applied limit, current reservations and waiting reads. A budget change applies without leaving the plot, and applicable budget-failed owners retry their current selection. This models simultaneous work, not just the size of an individual successful response. See native field commands and I/O reservations.

These values are implementation ceilings, not a promise that every computer can safely devote that amount to Studio. Frontend preparation and GPU textures have additional budgets. A renderer can be within its texture-size limit yet run out of combined native, JavaScript and GPU working memory. The system must retain the reason a representation was reduced or refused rather than silently labeling it full resolution.

The native netCDF API has process lifetime while each dataset file retains its shorter open/close lifetime. The loaded library remains available for internal process or thread cleanup callbacks; repeatedly unloading it after each read would risk callbacks into unloaded code. This lifetime policy does not retain every file or remove the reader's serialized native API access. The dynamic netCDF owner keeps this distinction explicit.

23.2 Cancellation is a state transition, not instantaneous preemption

Each native window read has a request identity, cancellation flag and phase. Cooperative loops check the flag during header reading, streaming and other interruptible work. A libzfp decode or native netCDF hyperslab call cannot be interrupted at an arbitrary instruction through this interface. Cancellation is checked at its boundaries; a cancelled result is then discarded. The receipt says cancellation was requested, not that an in-progress native call has already stopped.

The frontend has a complementary obligation: a late result from frame n must not replace frame $n + 1$ after the user scrubs forward. Request/generation identity and coherent scene publication solve a different problem from the native cancellation flag. Even a request too late to cancel can be prevented from publishing into the wrong view. The frame identity and scene generation owners make that distinction explicit.

JavaScript workers perform preparation and extraction outside the main UI thread. Transferable ArrayBuffers move ownership and detach the transferred buffer; SharedArrayBuffers share storage when the environment permits them. Nimbus's handoff protocol distinguishes a transferred render-only result from a retained scientific source and from a shared immutable dataset. The choice must preserve retries and cache users. Avoiding a structured-clone copy is useful, but does not establish that native IPC or GPU texture upload was zero-copy.

23.3 Launching and stopping a process

The native process runner checks the executable and configuration identities before spawning. It publishes immutable launch provenance and starts separate readers for stdout and stderr plus a waiter for process completion. This keeps a child blocked on a full output pipe from appearing to be a stalled numerical solver. The process exit state is distinct from a complete final frame or a verified checkpoint; readers must still inspect publication records.

Stopping is also owned. On applicable Unix paths, the runner requests graceful termination and escalates if the child does not exit within its grace period. It does not equate closing a plot with killing the simulation, and a failed process does not become a successful run merely because it emitted several valid early frames. Filesystem, process and application lifecycle operations must preserve their separate error states.

Remote execution stages an exact request on a user-configured SSH target, checks the remote executable's catalog and identities, and launches that process there. Receiving output has its own inventory, per-file hashes and completion/cancellation receipt. A cancelled transfer is not a complete local copy of the remote run. The remote compute owner and receive owner separate launching from verified retrieval.

This is remote process orchestration, not MPI domain decomposition. The current architecture does not automatically distribute one atmospheric grid across multiple networked hosts. SSH availability, remote CPU ISA, installed library ABI, GPU capabilities and local viewer support remain separate checks. The workstation being unavailable prevents new measurements there; it does not justify inferring their results from a successful local ARM build.

24 GPU execution and the path from a field to a screen pixel

Nimbus uses GPUs for two different jobs. The solver's Metal and Vulkan compute backends advance atmospheric state. Studio and the standalone Vulkan viewer turn an available state into an image. Changing a colormap, camera, opacity or rendering resolution does not change the simulated atmosphere. Conversely, selecting a solver backend does not select Studio's rendering API or give its webview direct access to the solver's GPU buffers.

The multi-GPU design section describes what it would take to partition the solver across devices. That future compute capability is separate from distributing rendering. Multiple graphics or compute queues are not evidence that a simulation is already sharded.

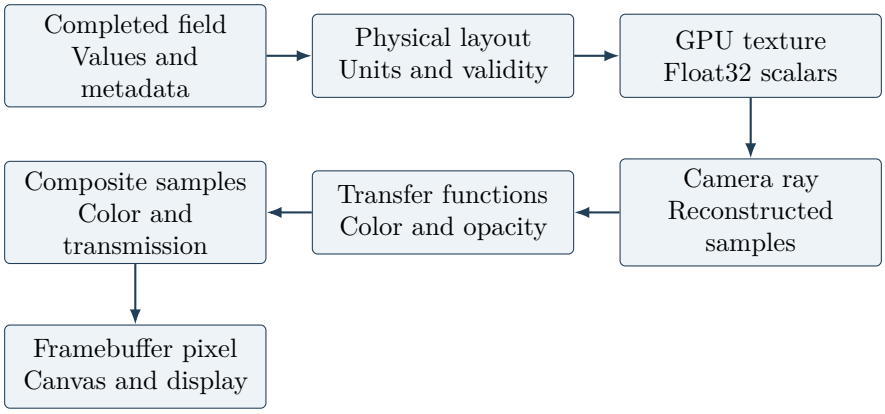


Figure 5: The volume-rendering path separates scientific values from their display interpretation. Camera, palette, opacity and lighting affect the image; they do not advance the atmospheric state. Geometry rendering uses a different path through vertices, triangles and rasterization.

24.1 Decode the field before interpreting its appearance

Studio's native Rust layer reads exported NPY, ZFP and netCDF fields, validates their metadata and returns binary arrays through Tauri. Large reads can request a bounded spatial window with

explicit origin, extent, stride and aggregation. The window protocol reports the source shape, returned shape, sample location, missing-data policy and byte accounting. Its output-byte limit is distinct from the amount of source data read: computing a block mean still visits the source cells represented by that block. See the native output reader and field-window protocol.

The numbers need physical coordinates and units. A dimension named z is not sufficient evidence of uniform vertical spacing, and closure/halo storage is not extra atmosphere to display. The reader checks valid physical ranges, scalar origins, staggering and terrain metadata. Windowed terrain data can carry double-precision physical centroids and cell volumes alongside float field values. Studio's Cartesian preparation resamples stretched or mapped geometry onto the regular lattice required by a volume image, retaining the interpretation of that reconstruction.

Units come from the field catalog rather than guesses based on the variable name. Derived-field recipes declare their inputs and required units; a saved native field takes precedence over a Studio-derived field with the same name. For example, temperature derived from theta and pressure remains a derived diagnostic with a recipe and dependencies. A horizontal-level anomaly is also a deliberate transformation, not raw temperature with a different palette. Unit labels, the derived-field registry and Cartesian preparation own these decisions.

Preparation then repacks values into VTK's x-fastest order. A common Nimbus array index is $(in_y + j)n_z + k$; the corresponding VTK scalar index is $i + n_x(j + n_y k)$. Setting dimensions without this repack can silently transpose a non-cubic storm. `vtkImageData` combines the scalar array with dimensions, origin and spacing; vectors have three components per point. Surface geometry instead uses `vtkPolyData` with points, connectivity and associated arrays. Neither object is a framebuffer or an already rendered picture. The VTK image adapter is the central layout boundary; Kitware's data-structure documentation describes the framework objects.

24.2 The actual Studio GPU APIs

Studio currently selects **WebGL for volume workloads**, including when the user prefers WebGPU. Its pinned vtk.js volume path does not provide the same scene-light, scattering, opacity-normal and close-camera behavior through WebGPU. This is an explicit workload policy, not CPU volume rendering. Scientific volume texture code requires an available WebGL2 context.

Eligible geometry scenes can select WebGPU. Adapter discovery is only the first check: Studio waits for the mounted `vtkWebGPURenderWindow` to initialize its device and confirms the active API before reporting it active. Missing support, initialization failure or device loss leads to the documented WebGL compatibility path and scene reconstruction. The authoritative policy is vtkBackend.ts; upstream WebGPU availability alone does not override it.

A vertex shader transforms mesh vertices into camera/clip coordinates and passes attributes onward. Rasterization determines the screen fragments covered by triangles or lines. A fragment shader computes color and opacity for those fragments. Depth and blending operations determine how their results enter the framebuffer. A compute shader instead runs explicit workgroups that read and write buffers or images without inherently drawing triangles. These are distinct pipeline roles in the Vulkan pipeline specification.

In Studio, worker-based preparation, marching-cubes extraction and streamline integration are CPU-side JavaScript work even though their results are drawn by GPU shaders. The WebGL volume ray marcher runs in a fragment shader, not a solver compute kernel. WebGPU geometry uses vtk.js's graphics pipeline with Nimbus's surface-lighting adaptation.

24.3 Upload a field, then look up a color

The VTK volume mapper needs a three-dimensional GPU texture. Nimbus's scalar texture owner explicitly disables half-float storage and uploads Float32 scalar data. It queries the context's 3D texture limit and float-linear filtering support. If hardware float-linear filtering is unavailable, a shader fetches the neighboring texels and performs trilinear interpolation explicitly. That fallback preserves the scalar precision choice rather than silently switching to a smaller texture format.

Texture dimensions and logical grid dimensions need not coincide. If one logical axis is too long but the total cell capacity fits, Nimbus packs the linear scalar sequence into a legal 3D texture and reconstructs its logical addresses in GLSL. The upload uses `texSubImage3D` for the occupied prefix and restores pixel-unpack state afterward. Capacity failures are reported; texture packing is not permission to discard cells. See texture packing and capability checks.

A color transfer function maps a scalar to RGB. For a range-based palette, the conceptual coordinate is

$$s = \mathrm{clamp}\left(\frac{q - q_{\min}}{q_{\max} - q_{\min}}, 0, 1\right).$$

The palette interpolates between color stops at that coordinate. This changes the display interpretation, not the source array. Signed fields can use a diverging palette to distinguish opposite signs. A constant field needs an explicit degenerate-range policy to avoid division by zero. Studio's dBZ palette uses physical dBZ anchors, so a specified dBZ value has the same palette color across runs rather than being stretched to each frame's extrema. VolumeRender3D applies these transfer functions and publishes their range for the legend.

Opacity is a separate transfer function. Two samples can have the same RGB but different visibility. Increasing opacity can obscure interior structure; changing the color range can make small variations appear dramatic. Neither operation increases the information resolved by the simulation. A scientific figure therefore needs its field, units, transformation, color range and opacity policy, not just the palette name.

24.4 How a volume becomes one pixel

For each covered screen fragment, the camera defines a ray through the volume. The shader intersects that ray with the volume's physical/display bounds and samples along the interior interval. A texture sample reconstructs a scalar from nearby grid values; transfer functions turn it into a color and opacity. Sampling between grid points smooths the displayed reconstruction but does not create new atmospheric resolution.

Opacity must account for distance. If a transfer-function opacity a_0 is defined over reference length ℓ_0, the corresponding opacity over a ray segment $\Delta\ell$ is

$$a = 1 - (1 - a_0)^{\Delta\ell/\ell_0}.$$

This is the constant-extinction Beer–Lambert relationship. Without the distance correction, taking twice as many samples would make the same column artificially more opaque. Nimbus also converts display distance back to physical distance: vertical exaggeration must not thicken a cloud optically simply by stretching its picture. Geometry spacing/origin, sample budgets and opacity correction

therefore change together. These are visualization extinction controls, not the atmospheric RRTMG optical coefficients. See physical opacity and optical-distance handling.

The front-to-back accumulation for premultiplied color C and opacity A is

$$C_{n+1} = C_n + (1 - A_n)a_n c_n,$$
$$A_{n+1} = A_n + (1 - A_n)a_n.$$

The remaining transmission weights everything behind the accumulated samples, including the background. Rays can stop when their interval ends or their remaining transparency is negligible. Nimbus's coherent sampling uses a stable pixel-dependent phase and shared sampling bins across bricks, weighting only each brick's owned segment. Otherwise arbitrary brick edges can change total opacity or create seams.

When several supported cloud/precipitation layers overlap, the joint renderer samples their local contributions together and combines optical depth. Drawing each entire volume as an independent translucent object can yield incorrect overlap ordering. The current joint owner supports up to four active scalar textures and owns their transforms, sample budgets and transfer functions. Coherent sampling and the joint fragment shader contain the actual accumulation rules.

Lighting modifies the displayed color using normals and scene lights. Surface normals come from geometry; lit volumes estimate gradients from reconstructed scalar or declared opacity information. Nimbus corrects gradient geometry for anisotropic spacing and uses a fixed world-space key/fill/rim rig rather than letting an automatic headlight change the material as the camera moves. Certain structural/cloud views also use volumetric shadow/scattering controls; generic scalar views can remain unlit. These are visual cues, not a measured photograph or the solver's solar-radiation calculation. The lighting owner records those choices.

Opaque geometry contributes depth, which the volume path receives through its z-buffer pass so integration can respect scene occlusion. Translucent geometry requires blending and ordering; Nimbus has an explicit ordered-renderer adaptation. The resulting color/alpha image resides in framebuffer attachments before the browser or native webview composites the canvas with controls and legends. Display color handling and final finite-bit pixels are another representation boundary. A screenshot pixel is not an invertible measurement of the original scalar after lighting, opacity, overlap and quantization.

24.5 Surfaces, contours, vectors and missing data

An isosurface answers a different question: where does a reconstructed scalar equal a chosen threshold? Marching cubes interpolates crossings inside cells and creates triangles. The GPU draws those triangles; it does not integrate every volume sample behind them. Surface color can be uniform or mapped from an attached scalar. A two-dimensional contour similarly traces a level set on a plane rather than representing integrated cloud opacity.

Vector glyphs display sampled direction and magnitude. Streamlines integrate a vector field at one time and then draw polylines or tubes; they are not automatically particle trajectories through a changing wind field. Their seed spacing, integration limits and interpolation determine what is shown. Studio's separate trajectory owner retains that distinction.

Missing values are not zeros. A finite substitute may be needed to keep a volume texture valid, but Studio retains a separate validity mask and samples it in the shader. The mask also protects normal estimation near gaps. Marching-cubes cells with missing corners are omitted, and streamline

support prevents a line from bridging an invalid region. The volume mask, finite marching-cubes adapter and finite streamline adapter implement different representations of the same missing-data obligation.

Level of detail is likewise explicit. Block means, extrema and point samples answer different questions; averaging dBZ requires its declared linear- reflectivity aggregation rather than an ordinary logarithmic mean. Terrain means use physical-volume information where the protocol declares it. Bricks, overview levels, temporary screen resolution and ray step size are distinct quality controls. Preparation carries resource budgets and source/window provenance, and generation ownership prevents a stale asynchronous result from replacing a newer field. Finer-looking interpolation cannot be labeled native resolution after a coarse source window has been selected.

24.6 Loading concurrency, cancellation and memory preferences

Studio separates the count of loading workers from the amount of memory those workers may hold. A worker limit bounds concurrent field reads and decoding. The shared native I/O limit accounts for a request's declared source working set plus twice its output allowance, covering the result and encoding workspace. The JavaScript scheduler also reserves its estimated read cost. These are admission bounds rather than measurements of the whole process: field caches, Python's WebAssembly heap, prepared geometry, browser copies and GPU allocations retain their own owners and limits.

The automatic native I/O allowance is one thirty-second of detected system RAM, bounded between 512 MiB and 4 GiB, and further limited by the manual ceiling. That ceiling is one quarter of RAM, bounded between 64 MiB and 16 GiB. If RAM is unknown, the default is 512 MiB and the manual ceiling is 2 GiB. A saved preference can select a smaller or larger allowance within the current computer's permitted range. These policies use detected capacity, not an ARM or x86 label; neither policy assumes that GPU memory is unified with RAM.

An individual request larger than the allowance is rejected with its required size. Ordinary contention instead waits on a native condition variable and checks cancellation. Reservations live with the worker and its completed payload until the response is handed back or discarded. Reducing the limit does not erase memory already owned by an active worker. Those workers finish or cancel, while later reads wait for capacity. The preference is published to the frontend only after native configuration succeeds. A delayed status reply cannot overwrite that newer configuration. Budget-related failed plot reads retry after a changed allowance is applied, without requiring the user to leave a volume view.

Prefetch belongs to the selected run, field and spatial window. Frame advance within that selection can join an existing future-frame read. Replacing or deselecting the field releases that selection's interest; a shared read is cancelled when it has no remaining consumers. Failed foreground reads do not start speculative work. Likewise, an empty isosurface releases its old geometry rather than merely hiding an actor while retaining its buffers. These lifetimes matter on both unified-memory and discrete-memory computers.

The first ordinary desktop session without a matching saved profile can measure loading workers after the interface becomes idle. The measurement generates a 2 MiB float32 NPY file and processes sixteen equal read-and-scan jobs per trial. Each candidate count receives a warmup and three trials; the median determines throughput. Candidate counts are powers of two through sixteen, bounded by the detected core count and including that bound. Studio chooses the smallest count within five percent of the best median throughput. Cancellation and a twenty second cooperative

deadline bound the work; an operating-system read already in progress must return before it can stop. Temporary files are removed when the measurement ends. Explicit worker preferences are preserved.

This is a cached-read and CPU-scan heuristic. It does not measure sustained storage throughput, netCDF or ZFP decoding, every plot's preparation cost, solver scaling, or GPU execution. The recommendation is associated with the CPU identity, core count and RAM reported by the native shell. It can be rerun after workload changes, and manual controls remain available. A successful short measurement is not a hardware qualification campaign.

Python array previews similarly separate representation from ownership. A large heatmap initially point-samples a bounded number of cells; zooming selects finer samples from the retained complete array. The color range remains tied to the full finite result, and export uses the original dtype and shape. Coordinate metadata uses explicit metre, radian, model-level or index units. Unknown physical geometry does not acquire metre labels through a nominal spacing.

24.7 The standalone Vulkan viewer

The native viewer has a separate path under `vulkan/src/rendering`. Its export reader loads and transposes field data, while a shared-memory input can provide live host-visible frames. These inputs still pass through the viewer's own normalization and texture upload; they do not alias a Studio texture or a solver compute buffer.

Its unipolar normalization uses percentile-clipped ranges. Signed data uses a centered encoding, with zero at 0.5 and opposite signs on opposite sides; a sampled cross-frame magnitude estimate is used when available, with a per-frame fallback. Nonfinite values are counted and replaced by neutral display values. This differs from Studio's explicit validity-mask behavior. The normalized intensity is therefore not a universal physical color scale. ExportDataset is the authority.

The viewer allocates R32 floating-point 3D images, checks required format and linear-filter support, and uploads each active field. A persistent two-slot ring holds mapped host-coherent staging buffers. CPU `memcpy` fills a free slot; image barriers make it a transfer destination; `vkCmdCopyBufferToImage` copies the data; a later barrier makes it shader-readable. Submission uses `vkQueueSubmit`, and slot fences protect reuse while earlier work is pending. The path remains explicit even on shared-memory hardware. See volume_backend.cpp.

The vertex shader emits one oversized fullscreen triangle. The fragment shader constructs camera rays, intersects the physically scaled box, samples the selected fields and accumulates transmittance. It uses a bounded ray-step count, anisotropy-aware gradients, display palettes, lighting and background sky. Optional natural/cinematic texture adds procedural appearance; it is not simulated small-scale cloud structure. This is a graphics-fragment ray marcher, not Vulkan hardware ray tracing or marching-cubes surface extraction.

The window renderer acquires a swapchain image, waits for the necessary frame ownership, submits drawing and presents it with `vkQueuePresentKHR`. It prefers an sRGB surface format when available. Screenshot capture has a separate image-to-buffer readback while the application still owns the image, with supported capture-format checks. Ordinary presentation does not require copying every pixel back to the CPU. WindowRenderer owns this final image-to-display boundary.

24.8 Memory sharing does not erase ownership or synchronization

Studio's `SharedArrayBuffer` path can avoid a JavaScript copy between workers and the main thread. When unavailable, it uses transferable copies with retained-source and retry rules. It also reuses immutable prepared datasets across camera changes. These are host handoff optimizations: "shared zero copy" does not mean the browser directly renders the solver's memory or that texture upload costs nothing. The handoff owner records the relevant stages and copies.

On discrete GPUs, host and device memory are separate pools; uploads and readbacks commonly traverse a bus. Unified memory can expose one compatible allocation to both processors, but capacity, visibility and execution ordering still matter. The Khronos memory guide describes these allocation distinctions. Nimbus's solver Vulkan backend chooses direct host-visible/device-local buffers only when its memory plan permits them; the staged path has separate host and device buffers. It uses buffer copies, barriers, fences and noncoherent flush/invalidate handling as required. Resident reuse additionally needs valid field identities and generations; reusing an allocation does not establish that its contents are current. The compute backend owns these rules independently of the viewer's image-upload ring.

Metal solver operators use declared buffer ownership, including shared storage paths. `newBufferWithBytes` and copying from a buffer's contents are real copies even on unified memory. Command completion still precedes consuming a result on the CPU. Host diagnostics or output may require materializing data that otherwise stays resident. Thus neither "Metal" nor "unified" guarantees a copy-free full timestep. The Metal advection owner shows both ordinary copy paths and resident reuse.

24.9 Rendering quality and numerical precision are separate contracts

Solver fields are commonly Float32, while selected compute kernels use compensated pairs, fused operations or explicit storage-rounding handling to preserve their numerical contracts. The mapped scalar helper and Vulkan storage helper are examples. These are scientific arithmetic choices, not image-quality settings. The Metal compiler's retained fast-math policy also belongs to the compute contract.

Rendering instead has its own precision, interpolation, sampling, texture and display constraints. A visually smooth cloud is not evidence of a conservative solver, and pixel differences do not by themselves prove a physics difference. Maintained tests separately cover scalar texture precision, missing-data rendering, joint volume composition, colormaps and scene lifecycle. Context/device loss disposes stale resources and rebuilds owned state before reporting a recovered frame. Capability fallback is visible; renderer availability is separate from native-only solver admission. These mechanisms explain behavior and limits, without establishing a universal performance claim or changing any scientific readiness status.

25 Performance models and the limits of measured evidence

25.1 Count work at the boundary that is actually limiting

The Roofline model relates attainable arithmetic throughput to both compute capacity and data movement. With operational intensity $I = W/B$, where W is floating-point work and B is bytes

transferred at a specified memory level, its elementary bound is

$$P \le \min(P_{\max}, \beta I), \qquad t \ge \max\left(\frac{W}{P_{\max}}, \frac{B}{\beta}\right).$$

Here β is sustained bandwidth at that same boundary. A memory-bound loop does not reach twice the performance merely because arithmetic width doubles. Williams, Waterman and Patterson develop this reasoning in the primary Roofline report. It is a useful bound and diagnostic model, not a prediction that every Nimbus loop reaches a roof.

For Nimbus, the choice of B matters. Counting bytes requested by source loads is different from measuring DRAM traffic after cache reuse. Host-to-device traffic is a different boundary again, and shader access to resident VRAM has another bandwidth. A CPU loop, a PCIe transfer, a GPU kernel and browser texture upload should not be placed under one unexplained bandwidth number. Extra validation or output scans can dominate a complete operation even when its inner stencil has a favorable arithmetic intensity.

The end-to-end consequence follows by separating an accelerated fraction f of runtime from everything else. If that fraction becomes s times faster, the idealized whole-program speedup is

$$S = \frac{1}{(1 - f) + f/s}.$$

This assumes the remaining work is unchanged and introduces no extra overhead. If a kernel is 5 percent of total time, even eliminating it entirely saves at most 5 percent under those assumptions. An actual change may add packing, allocation, synchronization or validation, so the measured result can be worse. The implication is to identify the costly owner before choosing an optimization, not to optimize the largest-looking loop by inspection.

25.2 Why a local speedup can disappear on another machine

The cross-architecture investigations have exposed three distinct questions: how much arithmetic an owner performs, how much data its consumers require, and when the hardware can execute that work. These questions remain useful without treating an early measurement on one host as a published benchmark or an acceptance threshold for a class of computers.

For CPU execution, increasing the worker count can stop helping once a shared cache, memory interface or synchronization point becomes limiting. SMT workers share physical resources, and thread placement can change which memory and cache paths an operator uses. Nimbus therefore exposes explicit worker and OpenMP placement controls and records the execution context. Its topology policy is a starting point. It does not encode a particular workstation's best observed count as a universal x86-64 setting.

Pressure-material reuse addresses a different problem: successive acoustic operations can need unchanged material coefficients while pressure itself changes. The current owners retain the immutable information under explicit state preconditions. That can remove repeated work in the pressure owner, but the complete step also includes transport, boundaries, microphysics, reports and synchronization. An owner-level improvement is a hypothesis about overall performance until the full workload is compared. Similarly, removing an upload is not sufficient if it adds another synchronization or shifts work to a less favorable point in the dependency chain.

On a device with separate memory, a staged acoustic command includes upload, execution, download and host publication. A timer around a transfer side can include waits for preceding work,

driver scheduling and visibility operations; it is not automatically an isolated measurement of PCIe bandwidth. The staged implementation keeps those phases explicit so they can be attributed and compared. It must preserve the same boundary and failure semantics as the shared-memory path before a transfer reduction is considered equivalent.

Power state, memory clocks, link state and competing work can vary between runs of an identical binary. A monitor sampled at a different cadence from the kernel trace cannot establish which individual call experienced a particular state. Nimbus's measurement workflow therefore keeps uncertainty visible and requires repeated, matched runs. The public repository does not publish a personal-device baseline here. Current code documents the implemented reuse, precision and memory-policy choices; broad ARM64, x86-64, NUMA and discrete-GPU performance conclusions still require the corresponding measurements.

25.3 A useful optimization experiment preserves the scientific job

The candidate and baseline must perform the same work: same configuration, duration, diagnostic cadence, output selection, memory policy, worker count and admission requirements. Reducing scientific reports or returning fewer fields can be a product-mode change, but is not an equivalent implementation speedup. Likewise, a native-required profile cannot obtain an acceptable timing by silently substituting a host operation.

Use an isolated owner measurement to locate cost, then evaluate the complete runtime path with warmup identified and repeated baseline/candidate runs. Alternating order helps reveal drift; a single before/after pair cannot separate a small change from varying load or operating state. Nested timers must state which intervals overlap. An acoustic sub-timer included in the dynamics timer is not an additional independent contribution to total time.

Correctness evidence accompanies the timing: accepted field bytes or declared numerical comparisons, checkpoints, relevant conservation ledgers, failures, threading and applicable backend paths. A source-only count of removed allocations is a hypothesis until the active runtime and retained dataflow confirm it. A portable test on one generic x86-64 build demonstrates that artifact's behavior there; it does not prove broad performance optimality.

26 What Studio's verification establishes and what remains bounded

A correct scientific viewer must do more than open a window. It must bind values to the intended coordinates, preserve units and missing-data semantics, show which frame actually finished, release abandoned work, and export the declared result. Generated analytical datasets can test these properties more decisively than a plausible storm image. Browser tests, native-reader tests, installed-package tests and sustained capacity measurements answer different questions. Passing one layer does not certify every other layer.

26.1 Local bounds are not yet a whole-application capacity guarantee

Studio has enforced limits for native I/O, caches, individual operations and device texture capabilities. The separate aggregate visualization governor only enables a qualified capacity policy when a matching accepted artifact is available. Without it, the aggregate policy is observe-only. See the capacity policy and native evidence admission. The adjustable I/O preference does not change that distinction or establish how much physical VRAM the whole application owns.

The version-2 packaged lifecycle report checks a narrower engineering contract. Two fresh scenes must render, delete their owned VTK objects, lose their retired graphics contexts, clear their canvas backing stores and release tracked resource claims. The report retains the whole-application RSS samples and signed changes from a common baseline. Those two samples cannot establish a memory plateau: browser compilation caches, garbage collection and native driver reclamation can continue after logical scene retirement. A passing lifecycle report therefore explicitly leaves the memory campaign pending and cannot produce an accepted aggregate-capacity artifact. Increasing memory across sustained cycles remains a capacity-investigation result, even when the bounded ownership checks pass.

The remaining capacity work needs the packaged browser engine, representative large outputs, simultaneous panes, whole-application resident memory, device allocations, opaque worker headroom and recovery under pressure. The maintained capacity artifact builder requires sustained profiles rather than a short interaction test. Above the full-read planning threshold, large-domain viewing requires an admissible bounded region or suitable multiresolution publication. Extra RAM alone does not supply missing overview data or remove a texture limit. These constraints belong in a local experiment plan because a simulation can finish successfully while unrestricted post-processing exceeds the viewer's working set.

26.2 Plotting, recording and Python have distinct supported scopes

The VTK backend policy currently uses WebGL for volume workloads. An available WebGPU geometry path does not establish equivalent volume materials, scattering or close-entry behavior. Software-GPU browser tests validate useful contracts but do not measure physical-device throughput or all driver and native-WebView combinations.

Recording requires an exact completed-frame identity and immutable capture semantics. The active Studio capture gate currently admits slice, vector, volume, isosurface, streamlines and supercell views. A plot being visible and interactive does not automatically make it eligible for recorded playback. Extending recording to another family requires its frame-readiness and export owner. MP4 export also depends on the native encoder path; a portable captured-view HTML export is not a freely rotatable replacement for the original three-dimensional data.

Python runs in a bounded WebAssembly environment. The Python protocol limits output arrays and materialization, and the array export client bounds supported tabular exports. A native preview budget is separate from the retained array and Python's internal allocations. Arbitrary user code can still exhaust the interpreter's heap; the restart path and the prerequisites for cooperative interrupts remain part of the user experience. Large produced files currently use a complete byte-array transfer, so general streaming analysis and export would require additional engineering.

Comparison also has a scientific admission boundary. The alignment owner requires compatible coordinates and units rather than silently interpolating arbitrary datasets onto one another. Automatic remapping or conversion would need an explicit operation, provenance and accuracy tests. A declared external derived-provider format likewise does not prove that every execution adapter is connected; the current native provider adapter is unavailable. These are functional limits to preserve visibly or complete deliberately, not reasons to fabricate plausible output.

26.3 A release needs its actual installed artifact

A browser harness imports source modules and generated fixtures. A desktop package additionally needs its exact native executable, runtime libraries, catalog, WebView, file dialogs and pro-

cess/export interfaces to agree. The tracked `catalog.runtime_cli_studio` readiness item in the runtime registry includes that agreement; a successful frontend build cannot close it. A compiler or locked dependency failure on another supported platform is a release blocker even when the viewer works on the developer's machine.

The next useful work is therefore concrete: complete reproducible package builds on each declared target, exercise ordinary native workflows and exact catalog selection, then validate sustained resource behavior on representative hardware. Lifecycle warnings should be resolved by checking the completeness of field, geometry and context identities and testing late completion and replacement. Blindly expanding hook dependencies can recreate an expensive scene repeatedly without fixing its ownership. Numerical review and scientific campaigns remain separate from all of these application checks.

27 Maintaining the ownership and provenance boundaries

Several inexpensive mistakes can invalidate a technically successful change. A new source writer can omit a residency transition; a reused scratch array can leave halos unreadably stale; a serializer can change canonical output while catalog pins remain old; or a cancellation path can release a reservation while a noninterruptible native call still owns its memory. These failures are best understood by asking who owns each resource, which generation its values describe, what constitutes completion, and which outputs must change together.

Maintained tests mirror those boundaries rather than relying on one aggregate smoke test. SIMD/alias and hardware-policy tests cover CPU representation and selection. Memory-plan and residency tests cover allocation and authority. Native backend tests cover actual dispatch, rejection and completion. Output and checkpoint tests cover publication and continuation. Studio tests cover preflight, decoding, geometry, missing data, resource lifetime and pixels. An analytic rendering fixture can expose an axis swap even when a storm image looks plausible; a hash mutation can expose a stale contract even when a UI control still renders successfully.

Generated files require their generator's exact contract. Formatting a generated shader string header or an exact-scope header can break its text verification without changing the apparent equation. Formatting a source file whose raw hash is sealed changes that source identity even if serialized output is unchanged. Such work should be isolated, regenerated and revalidated rather than mixed into an unrelated scientific or repository cleanup. Archived evidence must retain the identities it actually tested.

The repository's status vocabulary remains Engineering incomplete, Ready, Campaign validated, Externally reviewed and Fully qualified. Source inspection, bounded tests and this technical manual do not collapse those stages. In particular, complete bookkeeping of an energy change is not proof that the physical source is correct; a reproducible image is not proof of a physically validated storm; and a successful one-host benchmark is not a general hardware qualification.

28 Primary references and how they are used

The references below describe methods and platform semantics. Nimbus's actual mode choices, ABI, implementation limits and admission remain defined by the linked maintained source and tests. A paper's results are not measurements of the current Nimbus executable.

Compression method. Peter Lindstrom (2014), *Fixed-Rate Compressed Floating-Point Arrays*,

IEEE Transactions on Visualization and Computer Graphics, 20(12), 2674–2683. DOI: 10.1109/TVCG.2014.2346458. Primary paper PDF. Used for the block-compression design; Nimbus's containers and temporal predictor are separately specified in the output writer.

Compression modes and limits. ZFP 1.0.1 official documentation, Compression Modes and FAQ. Used to distinguish fixed rate, precision, accuracy and reversible operation, including finite-representation and pointwise-error caveats.

Performance model. Samuel Williams, Andrew Waterman and David Patterson, *Roofline: An Insightful Visual Performance Model for Floating-Point Programs and Multicore Architectures*, University of California, Berkeley Technical Report UCB/EECS-2008-134 (2008). Primary report PDF. Used as an operational-intensity bound, not as a fitted Nimbus performance curve.

CPU instruction sets and compilation. Arm, Arm C Language Extensions: Advanced SIMD intrinsics; Intel, Intel 64 and IA-32 Architectures Software Developer Manuals; GCC, x86 Options and x86 Built-in Functions. These define architectural and compiler semantics; emitted instructions and Nimbus's compiled helper branches determine what a particular build uses.

Thread placement. OpenMP Architecture Review Board.

OMP_PLACES and OMP_PROC_BIND. Used to distinguish worker count, places and affinity. Runtime support and observed placement remain necessary when interpreting a measurement.

Vulkan ownership and execution. Khronos Group, Memory Allocation, Synchronization Examples, and Vulkan Pipelines. Used for the distinction between allocation types, memory visibility, completion and graphics versus compute work.

Metal resources. Apple, Choosing a resource storage mode for Apple GPUs. Used for shared/private resource semantics. The current Nimbus backend's resource and completion choices are documented separately above.

Rendering framework. Kitware, VTK.js data structures. The pinned framework and Nimbus's workload policy, shader adaptations and permanent rendering tests define actual Studio support.

29 Repository boundaries

Location	Maintained responsibility
`include/`, `src/`	Solver interfaces, state, operators and orchestration
`vulkan/`, `metal/`	Native backend code, shaders and backend checks
`ui/`	Studio frontend and native shell; maintained tests live under `tests/ui/`
`configs/`	Public starting configurations
`tests/`	Numerical references, contracts and bounded fixtures
`scripts/`, `mk/`	Reproducible build, validation and operating tools

Location	Maintained responsibility
`docs/`	Current mathematical, architectural and operating explanations

Build products, downloaded data, simulation output and historical investigation records are not source dependencies. Generated directories are ignored by Git. Required runtime tables have explicit provisioning instructions in the build guide. A clean checkout must be able to rebuild and run the maintained checks without borrowing objects from an old experiment.